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A Nonconvolutional, Split-Field, Perfectly Matched Layer for Wave Propagation in Isotropic and Anisotropic Elastic Media: Stability Analysis

by Kristel C. Meza-Fajardo and Apostolos S. Papageorgiou

Abstract A nonconvolutional, split-field, perfectly matched layer, referred to as the multiaxial perfectly matched layer (M-PML), is proposed and implemented. The new formulation is obtained by generalizing the classical perfectly matched layer (PML; as originally proposed by Bérenger, [1994]) to a medium in which damping profiles are specified in more than one direction. Under the hypothesis of small damping and using an eigenvalue sensitivity analysis based on first derivatives, we propose a method to study the stability of the M-PML. With this method we demonstrate that the stability of the M-PML is related to the ratios of the specified damping profiles. Recognition of this fact leads to a general procedure for constructing robust, stable M-PML models for anisotropic media. It is also demonstrated that for any anisotropic medium the classical PML exhibits instabilities related to an eigenvalue with zero real part of multiplicity higher than one. Furthermore, we show that exponential growth due to eigenvalues with positive real part can be present in the classical PML for some orthotropic media. The effectiveness of the proposed M-PML and its advantages relative to the classical PML are demonstrated by constructing stable terminations for the aforementioned anisotropic media. The method of stability analysis is developed and demonstrated for two-dimensional elastodynamics problems, but its extension to three-dimensional configurations is straightforward.

Introduction

The perfectly matched layer (PML) model was introduced by Bérenger (1994) as a material boundary condition for electromagnetic wave propagation problems in unbounded domains. The model consists of an absorbing layer of finite width that surrounds the physical domain of interest so that all outgoing waves are damped out irrespective of their frequency and direction of propagation. The main feature of the PML is that, in the case of the continuum (i.e., before discretization), it does not generate reflections at the interface separating the absorbing layer and the physical medium (i.e., it is perfectly matched); however, a small reflection is always present after discretization. Being a material absorbing condition, the PML leads naturally to sparse systems; it is well suited for implementation in parallel computers and, consequently, is very attractive for numerical simulations. In its original version (i.e., as it was originally proposed by Bérenger [1994], we refer to this version as the classical PML), the field variables of the PML are split into nonphysical components so as to make it possible to incorporate in the mathematical formulation the desired absorption. This mathematical formulation of the PML has come to be known as the split formulation. Because of the effectiveness of the PML in absorbing the radiation exiting the physical domain, the extra memory storage required in order to accommodate the increased number of field variables is usually counterbalanced by the savings that are achieved by reducing the size of the physical domain. The satisfactory (in most cases) performance of the PML has resulted in a considerable amount of work toward its implementation in several wavelike problems (a concise yet comprehensive review of the subject is presented by Komatitsch and Martin [2007]), such as Maxwell's equations (e.g., Bérenger, 1996, 1998; Collino and Monk, 1998; Teixeira and Chew, 1998a), the linearized Euler equations (e.g., Hu, 1996; Hesthaven, 1998; Nataf, 2006), Helmholtz equations (e.g., Teixeira and Chew, 1998b; Harari *et al.*, 2000), and the elastic wave equations (e.g., Chew and Liu, 1996; Hastings *et al.*, 1996; Collino and Tsogka, 2001; Festa and Nielsen, 2003; Komatitsch and Tromp, 2003; Basu and Chopra, 2004).

Despite the general success of the PML in most applications, there are still some instances for which the performance of the PML does not meet expectations. In the case of elastic waves in isotropic media, it has been reported

(Festa *et al.*, 2005; Komatitsch and Martin, 2007) that large reflections are obtained for waves entering the PML at grazing incidence and also that instabilities appear in long (in time) simulations. Similar problems have been observed in the fields of electromagnetism (De Moerloose and Stuchly, 1995; Bérenger, 1998) and acoustics (Hu, 1996). In addition, Bécache *et al.* (2003) documented that exponentially growing solutions could appear in some models for anisotropic media. These problems have been intensively investigated in recent years, and a significant number of contributions on the subject has appeared in the published literature. For example, it was initially thought (Kuzuoglu and Mittra, 1996) that the source of the dynamical instability of the classical PML could be attributed to the fact the constitutive parameters did not satisfy causality, and an alternative causal frequency-dependent PML, known as the convolutional perfectly matched layer (C-PML), was proposed. Later, Teixeira and Chew (1999) showed that the original Cartesian PML parameters do indeed satisfy causality and that the aforementioned conclusion (regarding causality) was reached based on a technical error related to the Kramers–Kronig equations. Another approach was taken by Abarbanel and Gottlieb (1997) who suggested that because the split formulation is only weakly well posed, the PML solution diverged under small perturbations. This result motivated various investigators to develop unsplit formulations (Gedney, 1996; Zhao and Cangellaris, 1996; Ziolkowski, 1997; Abarbanel and Gottlieb, 1998; Basu and Chopra, 2004; Appelö and Kreiss, 2005). The unsplit formulations have the advantage of a more straightforward implementation in the various numerical schemes, although they also imply an additional computational cost, in some cases comparable to that of the split PML, due to the presence of either convolution operations, high-order time derivatives, or auxiliary variables. It was later found (Abarbanel *et al.*, 2002) that the unsplit formulations for Maxwell's equations were still susceptible to manifest slowly growing spurious solutions that eventually spread and pollute the physical domain. Subsequently, it was demonstrated (Bécache *et al.*, 2004) that the C-PML does not suffer from such instabilities. It was also reported that the C-PML improves the absorption of waves at grazing incidence (Roden and Gedney, 2000; Bérenger, 2002; Festa and Vilotte, 2005; Komatitsch and Martin, 2007). However, it should be noted that the C-PML behaves as an elastic medium at low frequencies, losing its absorptive capacity. Furthermore, at high frequencies C-PML reduces asymptotically to the PML, losing its stabilizing properties. In addition, Komatitsch and Martin (2007) observed that the C-PML does not solve all the instability problems of PML models for elastic anisotropic media reported by Bécache *et al.* (2003). More recently, Appelö and Kreiss (2005) succeeded in stabilizing some but not all those PML models for anisotropic media, by implementing a PML formulation obtained from a modification of the modal solutions of the elastic system (Hagstrom, 2003). Despite all the work done on the subject, a comprehensive mathematical analysis of the PML is not yet available, and the problem of developing a general method to construct stable PML terminations remains open.

In this article, we extend the formulation of the split-field PML to a more general absorptive medium, which we refer to as multiaxial perfectly matched layer (M-PML), for which damping profiles are specified along more than one direction. We also present a simple method to demonstrate that the stability of the system is related to the ratio of the damping profiles and thus provide a general procedure for constructing stable M-PML terminations for any model of anisotropic elastic media. The method provides a split-field M-PML for anisotropic elastodynamics that does not require convolution operations and remains efficient and stable for all frequencies and all directions of propagation. The analysis is presented for two-dimensional inplane (*P-SV*) wave propagation in elastic orthotropic media. However, extension of the results to three dimensions is straightforward. The generality of the method renders it suitable to study stability of PMLs for other wavelike problems (such as in acoustics and electromagnetism) as well. An analytical and numerical study of the accuracy of the M-PML for simulations using the spectral element method has been performed and will be reported in a future publication.

Split-Field Formulation of the Classical PML

The elastodynamics problem under no external forces is defined by Cauchy's equation of motion (also referred to as the momentum equation) and the generalized Hooke's law:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \mathbf{T}, \qquad \mathbf{T} = \mathbf{C}{:}\mathbf{E}, \tag{1}$$

where $\mathbf{u}(\mathbf{x}, t)$ is the displacement field, $\mathbf{x}$ is the position vector, $\mathbf{T}(\mathbf{x}, t)$ is the stress tensor, $\mathbf{E}(\mathbf{x}, t) = (\frac{1}{2})[\nabla \mathbf{u} + (\nabla \mathbf{u})^T]$ is the strain tensor, $\rho(\mathbf{x})$ is the mass density, and $\nabla = [\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z}]$. The preceding system of equations admits plane-wave solutions of the form

$$\mathbf{u} = \mathbf{A} \exp[-i(\mathbf{k} \cdot \mathbf{x} - \omega t)], \tag{2}$$

where $\mathbf{A}$ is the polarization vector, $i = \sqrt{-1}$, and ω is the circular frequency. The vector $\mathbf{k} = [k_x \quad k_y \quad k_z]$, known as the wave vector, gives the direction of propagation of the wavefront. Its amplitude $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2}$ is known as the wavenumber.

As stated by Komatitsch and Tromp (2003), the main idea of the classical PML is to construct a new wave equation that admits plane-wave solutions of the form

$$\mathbf{u} = \mathbf{A} \exp\left[-i(\mathbf{k} \cdot \mathbf{x} - \omega t) - \frac{\gamma_x}{\omega} k_x\right], \qquad \gamma_x(x) > 0, \tag{3}$$

where the additional factor $\exp(-\gamma_x k_x / \omega)$ modulates the amplitude of the wave resulting in waves that decay exponentially in the direction of increasing x. A general procedure

to derive PML formulations for linear wave propagation problems, known as coordinate stretching, was devised by interpreting the PML medium as an analytical continuation of the spatial variables of the original linear medium to a complex spatial variable domain (Chew and Weedon, 1994; Chew *et al.*, 1997). The complex space variable in the PML medium is then defined as follows:

$$\tilde{x} = x + \frac{\gamma_x}{i\omega}. \tag{4}$$

Then, the transformation (4) is used to derive a new operator $\nabla_{\tilde{\mathbf{x}}} = [\frac{\partial}{\partial \tilde{x}} \quad \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z}]$, where

$$\frac{\partial}{\partial \tilde{x}} = \frac{1}{s_x}\frac{\partial}{\partial x}, \qquad s_x = 1 + \frac{1}{i\omega}d_x, \qquad d_x = \frac{\partial \gamma_x}{\partial x}. \tag{5}$$

The function $d_x = d_x(x)$ is referred to as the damping profile and is specified to introduce attenuation, while s_x is referred to as the stretching factor. The new operator $\nabla_{\tilde{\mathbf{x}}}$ then replaces ∇ in system (1) to obtain the new wave equation, governing the motions of the PML medium, that we set out to construct. It is easy to verify that the plane wave expressed by equation (3) is obtained through substitution of equation (4) in the original expression (2), and thus equation (3) represents a solution of the new wave equation. Note that when $d_x = 0$ the operator $\nabla_{\tilde{\mathbf{x}}}$ reduces to that of the elastic medium. Bérenger's method then consists of placing the PML medium (which has a finite width by computational necessity) next to the boundary of the physical domain of interest (as shown in Fig. 1) so that the waves leaving the physical domain and entering the PML are damped out. To avoid reflections, the damping profile is selected so as to be zero at the physical domain–PML interface and smoothly increase across the PML width. The treatment of the regions where two or three PMLs overlap, the corner regions, is straightforward because its properties are just the superposition of the intersecting PMLs, that is, at the corner regions the coordinate stretching takes place in more than one direction. In a three-dimensional problem the new $\nabla_{\tilde{\mathbf{x}}}$ operator for a corner region becomes

$$\nabla_{\tilde{\mathbf{x}}} = \left[\frac{\partial}{\partial \tilde{x}} \quad \frac{\partial}{\partial \tilde{y}} \quad \frac{\partial}{\partial \tilde{z}}\right], \tag{6}$$

where

$$\frac{\partial(\cdot)}{\partial \tilde{x}_j} = \frac{1}{s_n}\frac{\partial(\cdot)}{\partial x_j}\delta_{jn}, \tag{7}$$

$$s_n = 1 + \frac{1}{i\omega}d_n, \tag{8}$$

$$d_n = \frac{\partial \gamma_n}{\partial x_j}\delta_{jn}, \tag{9}$$

where δ_{jn} denotes the Kronecker delta and we use the standard convention that repeated indices imply summation. In elastodynamics, one convenient way to implement the transformation expressed by equations (6)–(9) is to express system (1) as a first-order stress-velocity system in the frequency domain, using the velocity gradient tensor $\nabla\mathbf{v}$ instead of the strain tensor $\mathbf{E}$, and then making the substitution of the new operator $\nabla_{\tilde{\mathbf{x}}}$:

$$i\omega\rho v_i = \frac{1}{s_j}\frac{\partial T_{ji}}{\partial x_j}, \qquad i\omega T_{ij} = c_{ijkl}\left(\frac{1}{s_k}\frac{\partial v_l}{\partial x_k}\right). \tag{10}$$

In order to express the system in terms of the original operator ∇, velocities and stresses are split into components, each component being proportional to the individual terms on the right-hand side of equations (10) as follows:

$$i\omega\rho\sum_j v_i^{(j)} = \frac{1}{s_j}\frac{\partial T_{ji}}{\partial x_j}, \qquad i\omega\sum_k T_{ij}^{(k)} = c_{ijkl}\left(\frac{1}{s_k}\frac{\partial v_l}{\partial x_k}\right). \tag{11}$$

After splitting the velocity and stress tensor components, the stretching factors S_n are transferred to the left-hand side of the equations, and upon substitution of relation (8), the final time-domain system of split-field PML equations can be expressed as (Festa and Vilotte, 2005)

$$\begin{aligned} \rho\frac{\partial v_i^{(n)}}{\partial t} + \rho d_n v_i^{(n)} &= \frac{\partial T_{ij}}{\partial x_j}\delta_{jn}, \\ \frac{\partial T_{ij}^{(n)}}{\partial t} + d_n T_{ij}^{(n)} &= c_{ijkl}\frac{\partial v_l}{\partial x_k}\delta_{kn}. \end{aligned} \tag{12}$$

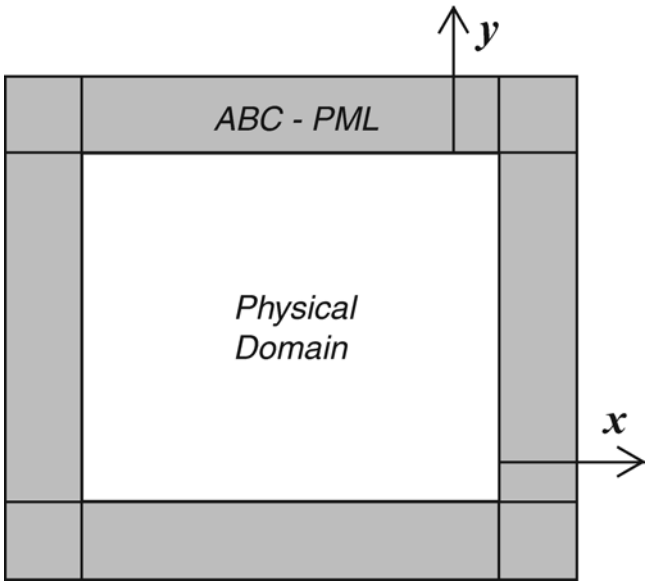


Figure 1. A physical domain, which extends beyond the boundaries displayed in the figure, is bounded by an Absorbing boundary condition (ABC) referred to as a PML. The two strips that are normal to the x axis are referred to as the vertical termination strips while the two strips that are normal to the y axis are referred to as the horizontal termination strips.

The M-PML Formulation

Suppose that the interface between the physical domain and the absorbing medium is a plane normal to the x axis at $x = 0$. The half-space corresponding to $x < 0$ is the physical domain, and the other half-space corresponding to $x > 0$ is the PML medium. In a classical PML, only one damping profile, d_x, is specified to be a function of the space variable x, whereas the other two profiles are set equal to zero:

$$d_x = d_x(x), \qquad d_y = 0, \qquad d_z = 0. \tag{13}$$

We propose to generalize the properties of the classical PML by selecting all three of the preceding damping profiles to be functions of the x coordinate:

$$d_x = d_x^{(x)}(x), \qquad d_y = d_y^{(x)}(x), \qquad d_z = d_z^{(x)}(x). \tag{14}$$

The additional damping profiles $d_y^{(x)}(x)$ and $d_z^{(x)}(x)$ are set to be proportional to $d_x^{(x)}(x)$:

$$\begin{aligned} &d_x = d_x^{(x)}(x), \qquad d_y = p^{(y/x)} d_x^{(x)}(x), \\ &d_z = p^{(z/x)} d_x^{(x)}(x), \qquad p^{(y/x)} = d_y^{(x)}/d_x^{(x)}, \\ &p^{(z/x)} = d_z^{(x)}/d_x^{(x)}. \end{aligned} \tag{15}$$

In view of the fact that $d_x^{(x)}(x)$ vanishes at $x = 0$, it becomes evident that all damping profiles vanish at $x = 0$, and consequently, the proposed M-PML medium retains the nonreflection characteristics of a PML. Then, the ratios of the damping profiles, namely, the constants $p^{(y/x)}$ and $p^{(z/x)}$, become parameters that allow fine-tuning of the stability of the M-PML medium.

The design of an M–PML medium is achieved by interpreting it not as a medium of complex space variables, but as a medium with complex wave vectors. Let us note that the solutions (3) in the case of the classical PML can alternatively be obtained by means of the following transformation:

$$\tilde{k}_x = k_x + \frac{1}{i\omega}\alpha_x k_x, \tag{16}$$

where $\alpha_x = \gamma_x/x$. Therefore, we can interpret the classical PML as a medium where only the component k_x of the wave vector is complex. In the case of M–PML, the generalization (14) of the damping profiles results in a medium with, in a sense, more regular properties, because all components of the wave vector are complex. The transformation to obtain the M–PML is then given by

$$\begin{aligned} &\tilde{k}_x = k_x + \frac{1}{i\omega}\alpha_x^{(x)} k_x, \qquad \tilde{k}_y = k_y + \frac{1}{i\omega}\alpha_y^{(x)} k_y, \\ &\tilde{k}_z = k_z + \frac{1}{i\omega}\alpha_z^{(x)} k_z. \end{aligned} \tag{17}$$

Substitution of these transformations in the product $\tilde{\mathbf{k}} \cdot \mathbf{x}$ gives

$$\begin{aligned} \tilde{\mathbf{k}} \cdot \mathbf{x} &= \tilde{k}_x x + \tilde{k}_y y + \tilde{k}_z z \\ &= k_x x + \frac{k_x}{i\omega}\alpha_x^{(x)} x + k_y y + \frac{k_y}{i\omega}\alpha_y^{(x)} y + \tilde{k}_z z + \frac{k_z}{i\omega}\alpha_z^{(x)} z. \end{aligned} \tag{18}$$

Then, the transformations that result from the use of the coordinate stretching approach are the following:

$$\begin{aligned} &\tilde{x} = x + \frac{1}{i\omega}\alpha_x^{(x)} x, \qquad \tilde{y} = y + \frac{1}{i\omega}\alpha_y^{(x)} y, \\ &\tilde{z} = z + \frac{1}{i\omega}\alpha_z^{(x)} z. \end{aligned} \tag{19}$$

The corresponding stretching factors s_n are

$$\begin{aligned} s_x &= \frac{\partial \tilde{x}}{\partial x} = 1 + \frac{1}{i\omega}\frac{\partial (x\alpha_x^{(x)})}{\partial x} = 1 + \frac{1}{i\omega}\frac{\partial (\gamma_x)}{\partial x} = 1 + \frac{1}{i\omega} d_x^{(x)}, \\ s_y &= \frac{\partial \tilde{y}}{\partial y} = 1 + \frac{1}{i\omega}\frac{\partial (y\alpha_y^{(x)})}{\partial y} = 1 + \frac{1}{i\omega}\alpha_y^{(x)}, \\ s_z &= \frac{\partial \tilde{z}}{\partial z} = 1 + \frac{1}{i\omega}\frac{\partial (z\alpha_z^{(x)})}{\partial z} = 1 + \frac{1}{i\omega}\alpha_z^{(x)}. \end{aligned} \tag{20}$$

Therefore, the damping profiles $d_x^{(x)}$, $d_y^{(x)}$, and $d_z^{(x)}$ are related to the functions $\alpha_x^{(x)}$, $\alpha_y^{(x)}$, and $\alpha_z^{(x)}$ as follows:

$$\begin{aligned} &\alpha_x^{(x)}(x) = \frac{\gamma_x(x)}{x} = \frac{1}{x}\int_0^x d_x^{(x)}(\xi)\, d\xi, \\ &\alpha_y^{(x)}(x) = d_y^{(x)}(x), \qquad \alpha_z^{(x)}(x) = d_z^{(x)}(x), \end{aligned} \tag{21}$$

while the plane-wave solutions in the M-PML take the following form:

$$\mathbf{u} = \mathbf{A} \exp(-\boldsymbol{\alpha} \cdot \mathbf{x}) \exp[-i(\mathbf{k} \cdot \mathbf{x} - \omega t)], \tag{22}$$

$$\boldsymbol{\alpha} = \frac{|\mathbf{k}|}{\omega}\begin{bmatrix} n_x\alpha_x^{(x)} & n_y\alpha_y^{(x)} & n_z\alpha_z^{(x)} \end{bmatrix}, \tag{23}$$

where n_x, n_y, and n_z are the direction cosines of the wave vector $\mathbf{k}$. As in the case of the classical PML, which clearly is a particular case of M-PML, the phase of the waves is maintained as they penetrate in the M-PML medium, and the modulating factor $\exp(-\boldsymbol{\alpha} \cdot \mathbf{x})$ introduces exponential decay that increases with increasing x. Furthermore, because the ratio $\frac{|\mathbf{k}|}{\omega}$ in equation (23) is equal to the reciprocal of the phase velocity, attenuation in the M-PML does not depend on frequency, and thus the M-PML is a nondispersive medium. The real-valued vector $\boldsymbol{\alpha}$ can be interpreted as the attenuation vector used for viscoelastic media, which gives the direction of maximum attenuation (Aki and Richards, 2002). Therefore, in the classical PML ($\alpha_y^{(x)} = 0$, $\alpha_z^{(x)} = 0$) the maximum attenuation is always in the direction of the x coordinate and is proportional to the k_x component of the wave vector. For waves entering the PML at grazing incidence, $k_x \sim 0$, and

thus the effect of attenuation is minimal. In the M-PML on the other hand, the components of $\boldsymbol{\alpha}$ are related to those of $\mathbf{k}$, that is, the direction of maximum attenuation is closely linked to the direction of propagation, and thus it is apparent that the absorption of waves entering the M-PML is more efficient, in particular, for the grazing incidence.

The treatment of the corner regions is similar to that of the classical PML, that is, the properties of the overlapping layers are superimposed, with the only difference being that now the damping profiles have additional parts that depend on other space coordinates:

$$\begin{aligned} d_x &= d_x^{(x)}(x) + d_x^{(y)}(y) + d_x^{(z)}(z), \\ d_y &= d_y^{(x)}(x) + d_y^{(y)}(y) + d_y^{(z)}(z), \\ d_z &= d_z^{(x)}(x) + d_z^{(y)}(y) + d_z^{(z)}(z). \end{aligned} \tag{24}$$

Because the generalization of the M-PML is on the damping profiles, implementation of the M-PML in codes that already use the classical PML requires only the modification of the damping profiles, and thus the necessary changes are minor.

Stability Analysis

In this section we perform a stability analysis of the M-PML for isotropic as well as orthotropic elastic media. Instabilities in PML implementations reported in the literature have been detected using different discretization methods, such as the finite differences method (Hu, 1996; Komatitsch and Martin, 2007), the finite element method (Bécache *et al.*, 2003), and the spectral element method (Festa *et al.*, 2005). Therefore, the stability analysis is carried out for the continuum; that is, we study the stability of the solutions of the partial differential equation that governs the motion of the M-PML.

M-PML Equations as a Linear Autonomous System

In this section the system of elastodynamic equations, expressed in their stress-velocity form, is rewritten in the time domain expressing the velocity gradient and stress tensors as column vectors and the elastic coefficient tensor as a matrix. The stability analysis is performed for M-PML terminations for a two-dimensional homogeneous elastic orthotropic medium whose axes of symmetry coincide with the x and y axes. The equations governing the inplane (*P-SV*) wave motion in an elastic medium can be expressed as a first-order system:

$$\begin{aligned} \rho\frac{\partial}{\partial t}\begin{bmatrix} v_x \\ v_y \end{bmatrix} &= \begin{bmatrix} \frac{\partial}{\partial x}T_{xx} + \frac{\partial}{\partial y}T_{yx} \\ \frac{\partial}{\partial x}T_{xy} + \frac{\partial}{\partial y}T_{yy} \end{bmatrix}, \\ \frac{\partial}{\partial t}\begin{bmatrix} T_{xx} \\ T_{yy} \\ T_{xy} \end{bmatrix} &= \begin{bmatrix} c_{11} & c_{12} & 0 & 0 \\ c_{12} & c_{22} & 0 & 0 \\ 0 & 0 & c_{33} & c_{33} \end{bmatrix} \begin{bmatrix} \frac{\partial v_x}{\partial x} \\ \frac{\partial v_y}{\partial y} \\ \frac{\partial v_y}{\partial x} \\ \frac{\partial v_x}{\partial y} \end{bmatrix}. \end{aligned} \tag{25}$$

The corresponding equations governing the motion of an isotropic elastic medium follow easily as a special case of the preceding equations by setting

$$c_{11} = c_{22} = \lambda + 2\mu, \qquad c_{12} = \lambda, \qquad c_{33} = \mu. \tag{26}$$

Following a procedure similar to that presented by Collino and Tsogka (2001), this system can be alternatively written as

$$\begin{aligned} \frac{\partial}{\partial t}\mathbf{v} &= \mathbf{D}^{(x)}\frac{\partial}{\partial x}\boldsymbol{\tau} + \mathbf{D}^{(y)}\frac{\partial}{\partial y}\boldsymbol{\tau}, \\ \frac{\partial}{\partial t}\boldsymbol{\tau} &= \mathbf{C}^{(x)}\frac{\partial}{\partial x}\mathbf{v} + \mathbf{C}^{(y)}\frac{\partial}{\partial y}\mathbf{v} \end{aligned} \tag{27}$$

with

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}, \qquad \boldsymbol{\tau} = \begin{bmatrix} T_{xx} \\ T_{yy} \\ T_{xy} \end{bmatrix}, \tag{28}$$

$$\mathbf{D}^{(x)} = \frac{1}{\rho}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{D}^{(y)} = \frac{1}{\rho}\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \tag{29}$$

$$\mathbf{C}^{(x)} = \begin{bmatrix} c_{11} & 0 \\ c_{12} & 0 \\ 0 & c_{33} \end{bmatrix}, \qquad \mathbf{C}^{(y)} = \begin{bmatrix} 0 & c_{12} \\ 0 & c_{22} \\ c_{33} & 0 \end{bmatrix} \tag{30}$$

for a nonzero mass density ρ. After applying transformations (19) and splitting the velocity and stress fields, we obtain the system of equations for the M-PML:

$$\begin{aligned} \frac{\partial}{\partial t}\mathbf{v}^{(x)} + d_x\mathbf{v}^{(x)} &= \mathbf{D}^{(x)}\frac{\partial}{\partial x}(\boldsymbol{\tau}^{(x)} + \boldsymbol{\tau}^{(y)}), \\ \frac{\partial}{\partial t}\mathbf{v}^{(y)} + d_y\mathbf{v}^{(y)} &= \mathbf{D}^{(y)}\frac{\partial}{\partial y}(\boldsymbol{\tau}^{(x)} + \boldsymbol{\tau}^{(y)}), \\ \frac{\partial}{\partial t}\boldsymbol{\tau}^{(x)} + d_x\boldsymbol{\tau}^{(x)} &= \mathbf{C}^{(x)}\frac{\partial}{\partial x}(\mathbf{v}^{(x)} + \mathbf{v}^{(y)}), \\ \frac{\partial}{\partial t}\boldsymbol{\tau}^{(y)} + d_y\boldsymbol{\tau}^{(y)} &= \mathbf{C}^{(y)}\frac{\partial}{\partial y}(\mathbf{v}^{(x)} + \mathbf{v}^{(y)}), \end{aligned} \tag{31}$$

where $d_x = d_x^{(x)}(x)$ and $d_y = d_y^{(x)}(x)$ are the damping profiles, defined and discussed in the previous section, and

$$\begin{aligned} \mathbf{v}^{(x)} &= \begin{bmatrix} v_x^{(x)} \\ v_y^{(x)} \end{bmatrix}, \qquad \mathbf{v}^{(y)} = \begin{bmatrix} v_x^{(y)} \\ v_y^{(y)} \end{bmatrix}, \\ \boldsymbol{\tau}^{(x)} &= \begin{bmatrix} T_{xx}^{(x)} \\ T_{yy}^{(x)} \\ T_{xy}^{(x)} \end{bmatrix}, \qquad \boldsymbol{\tau}^{(y)} = \begin{bmatrix} T_{xx}^{(y)} \\ T_{yy}^{(y)} \\ T_{xy}^{(y)} \end{bmatrix}. \end{aligned} \tag{32}$$

Assembling the preceding components of the split velocity and stress vectors in one vector, system (31) can be cast in the following form:

$$\frac{\partial\boldsymbol{\Psi}}{\partial t} = \mathbf{A}_x\frac{\partial\boldsymbol{\Psi}}{\partial x} + \mathbf{A}_y\frac{\partial\boldsymbol{\Psi}}{\partial y} - \mathbf{B}\boldsymbol{\Psi}, \tag{33}$$

where

$$\boldsymbol{\Psi} = \begin{bmatrix} T_{xx}^{(xx)} & T_{yy}^{(x)} & T_{xy}^{(x)} & T_{xx}^{(y)} & T_{xy}^{(y)} & T_{yy}^{(y)} & v_x^{(x)} & v_y^{(x)} & v_x^{(y)} & v_y^{(y)} \end{bmatrix}^T. \tag{34}$$

Matrices $\mathbf{A}_x$, $\mathbf{A}_y$, and $\mathbf{B}$ are block matrices having the following form:

$$\mathbf{A}_x = \begin{bmatrix} \mathbf{0}_{33} & \mathbf{0}_{33} & \mathbf{C}^{(x)} & \mathbf{C}^{(x)} \\ \mathbf{0}_{33} & \mathbf{0}_{33} & \mathbf{0}_{32} & \mathbf{0}_{32} \\ \mathbf{D}^{(x)} & \mathbf{D}^{(x)} & \mathbf{0}_{22} & \mathbf{0}_{22} \\ \mathbf{0}_{23} & \mathbf{0}_{23} & \mathbf{0}_{22} & \mathbf{0}_{22} \end{bmatrix},$$

$$\mathbf{A}_y = \begin{bmatrix} \mathbf{0}_{33} & \mathbf{0}_{33} & \mathbf{0}_{32} & \mathbf{0}_{32} \\ \mathbf{0}_{33} & \mathbf{0}_{33} & \mathbf{C}^{(y)} & \mathbf{C}^{(y)} \\ \mathbf{0}_{23} & \mathbf{0}_{23} & \mathbf{0}_{22} & \mathbf{0}_{22} \\ \mathbf{D}^{(y)} & \mathbf{D}^{(y)} & \mathbf{0}_{22} & \mathbf{0}_{22} \end{bmatrix}, \tag{35}$$

$$\mathbf{B} = \begin{bmatrix} d_x\mathbf{I}_3 & \mathbf{0}_{33} & \mathbf{0}_{32} & \mathbf{0}_{32} \\ \mathbf{0}_{33} & d_y\mathbf{I}_3 & \mathbf{0}_{32} & \mathbf{0}_{32} \\ \mathbf{0}_{23} & \mathbf{0}_{23} & d_x\mathbf{I}_2 & \mathbf{0}_{22} \\ \mathbf{0}_{23} & \mathbf{0}_{23} & \mathbf{0}_{22} & d_y\mathbf{I}_2 \end{bmatrix}, \tag{36}$$

where $\mathbf{0}_{nm}$ is the $n \times m$ zero matrix and $\mathbf{I}_n$ is the $n \times n$ identity matrix. The diagonal matrix $\mathbf{B}$ is referred to as the damping matrix. Furthermore, it is assumed that the mass density ρ and damping profiles d_x and d_y have constant values (in this case we refer to d_x and d_y as damping coefficients) so as to make Fourier analysis in space tractable. Following Bécache *et al.* (2003), we link the results of the analysis performed for a medium with constant coefficients to those of a medium with variable coefficients by invoking the frozen coefficient technique expounded in Kreiss and Lorenz (1989). In the system that results by taking the Fourier transform (with respect to the spatial variables) of equation (33), partial differentiation with respect to the x and y spatial coordinates is represented by the factors ik_x and ik_y, respectively. The transformed set of equations may be written as a 10×10 system that takes the following form:

$$\frac{\partial \mathbf{U}}{\partial t} = \mathbf{A}\mathbf{U}, \tag{37}$$

where

$$\mathbf{U}(k_x, k_y, t) = \int_{-\infty}^{+\infty} \Psi(x, y, t) \exp(ik_x x + ik_y y)\, dx\, dy \tag{38}$$

and $\mathbf{A}$ is a block matrix given by

$$\mathbf{A} = \begin{bmatrix} -d_x\mathbf{I}_3 & \mathbf{0}_{33} & ik_x\mathbf{C}^{(x)} & ik_x\mathbf{C}^{(x)} \\ \mathbf{0}_{33} & -d_y\mathbf{I}_3 & ik_y\mathbf{C}^{(y)} & ik_y\mathbf{C}^{(y)} \\ ik_x\mathbf{D}^{(x)} & ik_x\mathbf{D}^{(x)} & -d_x\mathbf{I}_2 & \mathbf{0}_{22} \\ ik_y\mathbf{D}^{(y)} & ik_y\mathbf{D}^{(y)} & \mathbf{0}_{22} & -d_y\mathbf{I}_2 \end{bmatrix}. \tag{39}$$

Because the coefficients of matrix $\mathbf{A}$ do not depend on the time variable, system (37) is a time-invariant or autonomous system. We can then apply the classical spectral stability criteria provided by the theory of linear dynamical systems (e.g., Hinrichsen and Pritchard, 2005; Velhurst, 2006). This theory studies the stability of linear autonomous systems by investigating the behavior of the solutions close to the equilibrium points. An equilibrium point is a solution that is time independent (i.e., a solution that satisfies $\partial\mathbf{U}/\partial t = 0$). Note that if matrix $\mathbf{A}$ in (37) is nonsingular, the only equilibrium point of the autonomous system is the origin $\mathbf{U}(t) = 0$. Then, one question that has to be addressed is whether solutions that start to evolve at time $t = t_0$ in a neighborhood of the origin will stay in this neighborhood for $t > t_0$. A stronger type of stability would require that solutions not only stay in a neighborhood of the origin but also converge to it with time. This is the notion of asymptotic stability. An equilibrium point is called asymptotically stable if all solutions $\mathbf{U} \in \mathbb{R}^n$ approach the equilibrium point as $t \to \infty$. Furthermore, as briefly outlined in the Appendix, the equilibrium solution at the origin $\mathbf{U}(t) = \mathbf{0}$ is asymptotically stable if all the eigenvalues $\{\sigma_i\}$ of $\mathbf{A}$ have negative real parts. For the autonomous system (37), this means that all solutions $\mathbf{U}(t)$ are composed of transients (i.e., terms that exponentially decay to zero). When at least one eigenvalue σ_i has a positive or zero real part, a nontransient solution exists. Because an absorbing layer like the M-PML is designed to dissipate all energy exiting the physical domain, asymptotic stability for the M-PML is desired. Thus, for the purposes of this work, we consider system (37) as stable if all the eigenvalues $\{\sigma_i\}$ have negative real parts; otherwise, the system is unstable. In the next section we propose a simple procedure to determine whether there are eigenvalues σ_i of $\mathbf{A}$ with a positive or zero real part.

Eigenvalue Sensitivity Analysis

When the original elastic unsplit 5×5 system (27) is expressed as an autonomous system,

$$\frac{\partial \mathbf{U}^u}{\partial t} = \mathbf{A}^u\mathbf{U}^u, \tag{40}$$

where the superscript u stands for unsplit and undamped, $\mathbf{U}^u = \begin{bmatrix} T_{xx} & T_{yy} & T_{xy} & v_x & v_y \end{bmatrix}$, and the matrix $\mathbf{A}^u$ takes the form

$$\mathbf{A}^u = \begin{bmatrix} 0 & 0 & 0 & ik_x c_{11} & ik_y c_{12} \\ 0 & 0 & 0 & ik_x c_{12} & ik_y c_{22} \\ 0 & 0 & 0 & ik_y c_{33} & ik_x c_{33} \\ ik_x \frac{1}{\rho} & 0 & ik_y \frac{1}{\rho} & 0 & 0 \\ 0 & ik_y \frac{1}{\rho} & ik_x \frac{1}{\rho} & 0 & 0 \end{bmatrix}, \tag{41}$$

the associated eigenvalue problem is given by

$$\sigma_i^u \mathbf{V}_i^u = \mathbf{A}^u \mathbf{V}_i^u, \tag{42}$$

where σ_i^u and $\mathbf{V}_i^u$ are the ith eigenvalue and the ith eigenvector of the system, respectively. The eigenvalues corresponding to system (40) are then easily obtained:

$$\sigma_i^u = \left\{ \pm \frac{i}{\sqrt{2}} \sqrt{(c_4 k_y^2 + c_3 k_x^2) + \sqrt{(c_4 k_y^2 + c_3 k_x^2)^2 - 4(c_1 k_x^4 + c_2 k_y^4 + c_5 k_x^2 k_y^2)}}, \right.$$
$$\left. \pm \frac{i}{\sqrt{2}} \sqrt{(c_4 k_y^2 + c_3 k_x^2) - \sqrt{(c_4 k_y^2 + c_3 k_x^2)^2 - 4(c_1 k_x^4 + c_2 k_y^4 + c_5 k_x^2 k_y^2)}}, \quad 0 \right\}, \tag{43}$$

where

$$c_1 = \frac{c_{33} c_{11}}{\rho^2}, \qquad c_2 = \frac{c_{33} c_{22}}{\rho^2}, \qquad c_3 = \frac{c_{11} + c_{33}}{\rho},$$
$$c_4 = \frac{c_{22} + c_{33}}{\rho}, \qquad c_5 = \frac{c_{11} c_{22} - c_{12}^2 - 2 c_{12} c_{33}}{\rho^2}. \tag{44}$$

We note that for an isotropic medium, eigenvalues (43) take the following simple form:

$$\sigma_i^u = \left\{ \pm i \sqrt{\frac{\lambda + 2\mu}{\rho}(k_y^2 + k_x^2)}, \quad \pm i \sqrt{\frac{\mu}{\rho}(k_y^2 + k_x^2)}, \quad 0 \right\}. \tag{45}$$

The four nonzero eigenvalues in (43) are then identical to the eigenvalues obtained for the second-order (i.e., displacement) formulation, and they correspond to the propagating quasi-*P* and quasi-*S* modes (e.g., Slawinski, 2003). Because of the symmetry of the elastic coefficient tensor $\mathbf{C}$, those four eigenvalues remain purely imaginary for all k_x and k_y (e.g., Slawinski, 2003). It is important to remark that when the two-dimensional elastodynamics equations for inplane motion are written as a first-order system (as in the present case), an additional spurious zero eigenvalue (corresponding to a nonpropagating/rigid-body mode) is introduced. This eigenvalue should be considered in the mathematical analysis of PML models, if their construction is based on the first-order (i.e., stress velocity) formulation. In the case of the split elastic (i.e., undamped) system, the constant coefficient matrix $\mathbf{A}^e$ (e stands for elastic)

$$\mathbf{A}^e = \begin{bmatrix} \mathbf{0}_{33} & \mathbf{0}_{33} & ik_x \mathbf{C}^{(x)} & ik_x \mathbf{C}^{(x)} \\ \mathbf{0}_{33} & \mathbf{0}_{33} & ik_y \mathbf{C}^{(y)} & ik_y \mathbf{C}^{(y)} \\ ik_x \mathbf{D}^{(x)} & ik_x \mathbf{D}^{(x)} & \mathbf{0}_{22} & \mathbf{0}_{22} \\ ik_y \mathbf{D}^{(y)} & ik_y \mathbf{D}^{(y)} & \mathbf{0}_{22} & \mathbf{0}_{22} \end{bmatrix} \tag{46}$$

has the following set of eigenvalues:

$$\sigma_i^e = \left\{ \pm \frac{i}{\sqrt{2}} \sqrt{(c_4 k_y^2 + c_3 k_x^2) + \sqrt{(c_4 k_y^2 + c_3 k_x^2)^2 - 4(c_1 k_x^4 + c_2 k_y^4 + c_5 k_x^2 k_y^2)}}, \right.$$
$$\left. \pm \frac{i}{\sqrt{2}} \sqrt{(c_4 k_y^2 + c_3 k_x^2) - \sqrt{(c_4 k_y^2 + c_3 k_x^2)^2 - 4(c_1 k_x^4 + c_2 k_y^4 + c_5 k_x^2 k_y^2)}}, \quad 0^{(6)} \right\}, \tag{47}$$

where the exponent in the zero eigenvalue denotes its multiplicity. Clearly, five of the eigenvalues are identical to those given in (43) for the unsplit (and undamped) system. The additional five eigenvalues that are equal to zero are introduced when the velocity and stress fields are split. On the other hand, it is evident from equation (47) that the real parts of all eigenvalues of the elastic system are zero. It is expected that the introduction of damping would have the effect of moving the eigenvalues toward the negative half complex plane (i.e., the half complex plane corresponding to the negative real semiaxis), and thus the resulting system would be

asymptotically stable. However, in view of the fact that the coefficient matrices in systems (37) and (40) are asymmetric, the presence of the damping matrix may have a destabilizing effect on the system. A physical explanation for this phenomenon has been given by several researchers (e.g., Semler *et al.*, 1998; Gallina, 2003). These investigators attribute the appearance of instabilities to modifications that the modes of vibration of the dynamical system undergo due to introduction of damping. Such modifications of the modes of vibration could cause the work done by external forces to increase. If the work done by external forces exceeds the energy dissipated by damping, the system becomes unstable.

In the case of the M-PML, the characteristic equation of matrix **A** as in (39) is given by

$$(\sigma + d_x)(\sigma + d_y)(\sigma^8 + q_7\sigma^7 + q_6\sigma^6 + q_5\sigma^5 + q_4\sigma^4 + q_3\sigma^3 + q_2\sigma^2 + q_1\sigma + q_0) = 0, \tag{48}$$

where

$$\begin{aligned}
q_0 &= d_x^4 d_y^4 + c_1 k_x^4 d_y^4 + c_2 d_x^4 k_y^4 + c_3 d_x^2 d_y^4 k_x^2 + c_4 d_x^4 d_y^2 k_y^2 \\
&\quad + c_5 d_x^2 d_y^2 k_x^2 k_y^2, \\
q_1 &= 4 d_x^3 d_y^3 (d_x + d_y) + 4(c_1 k_x^4 d_y^3 + c_2 d_x^3 k_y^4) \\
&\quad + 2c_3 d_x d_y^3 k_x^2 (2d_x + d_y) + 2c_4 d_x^3 d_y k_y^2 (d_x + 2d_y) \\
&\quad + 2c_5 d_x d_y k_x^2 k_y^2 (d_x + d_y), \\
q_2 &= 2 d_x^2 d_y^2 (3d_x^2 + 3d_y^2 + 8 d_x d_y) + 6(c_1 d_y^2 k_x^4 + c_2 d_x^2 k_y^4) \\
&\quad + c_3 d_y^2 k_x^2 (6d_x^2 + d_y^2 + 8 d_x d_y) \\
&\quad + c_4 d_x^2 k_y^2 (d_x^2 + 6d_y^2 + 8 d_x d_y) \\
&\quad + c_5 d_x^2 k_y^2 (d_x^2 + d_y^2 + 4 d_x d_y), \\
q_3 &= 4 d_x d_y (d_x + d_y)(d_x^2 + d_y^2 + 5 d_x d_y) \\
&\quad + 4(c_1 d_y k_x^4 + c_2 d_x k_y^4) + 4c_3 d_y k_x^2 (d_x^2 + d_y^2 + 3 d_x d_y) \\
&\quad + 4c_4 d_x k_y^2 (d_x^2 + d_y^2) + 12 c_4 d_x^2 d_y k_y^2 \\
&\quad + 2c_5 k_x^2 k_y^2 (d_x + d_y), \\
q_4 &= c_1 k_x^4 + c_2 k_y^4 + c_5 k_x^2 k_y^2 + 4 d_x d_y (4d_x^2 + 4d_y^2 + 9 d_x d_y) \\
&\quad + d_x^4 + d_y^4 + c_3 k_x^2 (d_x^2 + 6d_y^2 + 8 d_x d_y) \\
&\quad + c_4 k_y^2 (6d_x^2 + d_y^2 + 8 d_x d_y), \\
q_5 &= 4(d_x + d_y)(d_x^2 + d_y^2 + 5 d_x d_y) + 2c_3 k_x^2 (d_x + 2d_y) \\
&\quad + 2c_4 k_y^2 (2d_x + d_y), \\
q_6 &= 6d_x^2 + 16 d_x d_y + 6d_y^2 + c_3 k_x^2 + c_4 k_y^2, \\
q_7 &= 4(d_x + d_y).
\end{aligned} \tag{49}$$

Two eigenvalues are obtained by inspection:

$$\sigma_1 = -d_x, \qquad \sigma_2 = -d_y. \tag{50}$$

This result shows that, when damping is introduced, two of the six eigenvalues of the undamped split-field system that are equal to zero move to the left half of the complex plane. However, in the case of the classical PML, one of the two damping coefficients (50) is zero, and thus, with this simple result it becomes apparent that the classical PML medium is not asymptotically stable. On the other hand, there are no general closed form expressions for the remaining eight eigenvalues, because they are the roots of the polynomial of degree eight in (48). However, we recall that in order to characterize the stability of the M-PML system, we need to know only the sign of the real part of the eigenvalues. Furthermore, we recall that all eigenvalues of the undamped system have zero real parts and therefore lie on the imaginary axis of the complex plane. Therefore, all we need to know is the direction toward which the eigenvalues of the undamped system would tend to move as soon as we introduce damping into the system. In order to foresee which direction on the complex plane those eight eigenvalues would move by the introduction of damping, we perform an eigenvalue sensitivity analysis (Adhikari and Friswell, 2001; Gallina, 2003). The method consists of finding the first derivatives of the eigenvalues with respect to the damping coefficient evaluated at the point where that damping coefficient is zero. If the amount of damping introduced is small, the derivative will indicate toward which side of the complex plane, relative to the imaginary axis, the eigenvalue will move. The procedure starts by expressing the damping matrix **B** as a product of a matrix $\tilde{\mathbf{B}}$ and the damping coefficient $d_x^{(x)}$, namely,

$$\mathbf{B} = d_x^{(x)} \tilde{\mathbf{B}} = d_x^{(x)} \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{33} & \mathbf{0}_{32} & \mathbf{0}_{32} \\ \mathbf{0}_{33} & p^{(y/x)}\mathbf{I}_3 & \mathbf{0}_{32} & \mathbf{0}_{32} \\ \mathbf{0}_{23} & \mathbf{0}_{23} & \mathbf{I}_2 & \mathbf{0}_{22} \\ \mathbf{0}_{23} & \mathbf{0}_{23} & \mathbf{0}_{22} & p^{(y/x)}\mathbf{I}_2 \end{bmatrix}, \tag{51}$$

where the definitions/notation established in (15) have been used. Note that if $d_x^{(x)} = 0$, equation (33), and therefore matrix **A** in (39), reduce to the case of the undamped split-field system. As $d_x^{(x)}$ increases, the eigenvalues will tend to move either toward the left, negative half or toward the right, positive half complex plane. If the values of $d_x^{(x)}$ are small, a simple way to detect the direction of motion of the ith eigenvalue is to evaluate its first derivative with respect to $d_x^{(x)}$ when $d_x^{(x)} = 0$, namely,

$$\left.\frac{d\sigma_i}{d d_x^{(x)}}\right|_{d_x^{(x)}=0}. \tag{52}$$

Because in the undamped case all eigenvalues σ_i^e have zero real part, if $\mathrm{Re}(d\sigma_i/d d_x^{(x)}|_{d_x^{(x)}=0}) > 0$ for all i, a small $d_x^{(x)}$ will induce motion of the eigenvalues toward the positive half complex plane, causing the system to become unstable. While if

$$\mathrm{Re}\left(\left.\frac{d\sigma_i}{d d_x^{(x)}}\right|_{d_x^{(x)}=0}\right) < 0 \quad \forall\ i, \tag{53}$$

a small $d_x^{(x)}$ will induce motion of the eigenvalues toward the negative half complex plane, causing the system to become asymptotically stable.

In the analyses presented by Adhikari and Friswell (2001), the derivatives are obtained from expressions requiring the right and left eigenvectors of the undamped system. Here we adopt a different approach and derive exact expressions for the derivatives by applying implicit differentiation to the characteristic equation. In fact, because two eigenvalues have already been explicitly obtained, the procedure is applied only to the polynomial of degree eight given in (48). The derivative of the mth root of that eighth degree polynomial with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$ obtained by implicit differentiation is given by (Meza-Fajardo, 2007)

$$\begin{aligned}\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} &= -\frac{1}{D}\{2(1+p^{(y/x)})(\sigma_m^e)^4 \\ &+ [c_3k_x^2(1+2p^{(y/x)}) + c_4k_y^2(2+p^{(y/x)})](\sigma_m^e)^2 \\ &+ 2(c_1p^{(y/x)}k_x^4 + c_2k_y^4) + c_5k_x^2k_y^2(1+p^{(y/x)})\}, \end{aligned} \quad (54)$$

where

$$\begin{aligned}D = 4(\sigma_m^e)^4 &+ 3(c_3k_x^2 + c_4k_y^2)(\sigma_m^e)^2 \\ &+ 2(c_1k_x^4 + c_2k_y^4 + c_5k_x^2k_y^2)\end{aligned} \quad (55)$$

and the definitions (15) have been used. Because the derivative is evaluated at $d_x^{(x)} = 0$, the eigenvalue that appears on the right-hand side of the preceding equation is the eigenvalue σ_m^e of the undamped system. Four out of the eight eigenvalues $\{\sigma_m^e : m = 1, 2, \ldots, 8\}$ are zero and the other four that are nonzero are given in closed form in (47). Furthermore, because all those eigenvalues are either zero or purely imaginary, the factors $(\sigma_m^e)^4$ and $(\sigma_m^e)^2$ are real, and consequently derivative (54) is also real, that is,

$$\mathrm{Re}\left(\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0}\right) = \left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0}. \quad (56)$$

Now, because the M-PML medium is by construction anisotropic, the behavior of a strip perpendicular to the x axis (referred to as the vertical termination strip) is not the same as that of a strip perpendicular to the y axis (referred to as the horizontal termination strip). In the case of the horizontal termination strip, the damping profiles are defined as

$$\begin{aligned}&d_x = d_x^{(y)} = p^{(x/y)}d_y^{(y)}(y), \qquad d_y = d_y^{(y)}(y), \\ &p^{(x/y)} = d_x^{(y)}/d_y^{(y)}, \end{aligned} \quad (57)$$

and thus the derivatives should be computed with respect to the damping coefficient $d_y^{(y)}$ and evaluated at $d_y^{(y)} = 0$. Again, the mth eigenderivative is obtained by implicit differentiation of the polynomial in (48) (Meza-Fajardo, 2007)

$$\begin{aligned}\left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} &= -\frac{1}{D}\{2(1+p^{(x/y)})(\sigma_m^e)^4 \\ &+ [c_3k_x^2(2+p^{(x/y)}) + c_4k_y^2(1+2p^{(x/y)})](\sigma_m^e)^2 \\ &+ 2(c_1k_x^4 + c_2p^{(x/y)}k_y^4) + c_5k_x^2k_y^2(1+p^{(x/y)})\} \end{aligned} \quad (58)$$

with D as in (55). It becomes clear that the eigenderivatives (54) and (58) are determined by the elastic constants, the direction of propagation, and by the ratios $p^{(y/x)}$ and $p^{(x/y)}$, respectively. This is consistent with the conclusions of Herrmann *et al.* (1966) and Bolotin and Zhinzher (1969) who showed that the destabilizing effect induced by the introduction of a slight amount of damping on dynamic systems can be related to the ratio of the damping coefficients. Furthermore, because it is always possible to express the eigenvalues in terms of the wavenumber, equations (54) and (58) can be recast in such a form as to reveal that they are independent of wavenumber $|\mathbf{k}|$ and, consequently, independent of frequency, as follows:

$$\begin{aligned}\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} &= -\frac{1}{\tilde{D}}\{2(1+p^{(y/x)})(\tilde{\sigma}_m^e)^4 \\ &+ [c_3n_x^2(1+2p^{(y/x)}) + c_4n_y^2(2+p^{(y/x)})](\tilde{\sigma}_m^e)^2 \\ &+ 2(c_1p^{(y/x)}n_x^4 + c_2n_y^4) + c_5n_x^2n_y^2(1+p^{(x/y)})\}, \end{aligned} \quad (59)$$

$$\begin{aligned}\left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} &= -\frac{1}{\tilde{D}}\{2(1+p^{(x/y)})(\tilde{\sigma}_m^e)^4 \\ &+ [c_3n_x^2(2+p^{(x/y)}) + c_4n_y^2(1+2p^{(x/y)})](\tilde{\sigma}_m^e)^2 \\ &+ 2(c_1n_x^4 + c_2p^{(x/y)}n_y^4) + c_5n_x^2n_y^2(1+p^{(x/y)})\} \end{aligned} \quad (60)$$

with

$$\begin{aligned}\tilde{D} = 4(\tilde{\sigma}_m^e)^4 &+ 3(c_3n_x^2 + c_4n_y^2)(\tilde{\sigma}_m^e)^2 \\ &+ 2(c_1n_x^4 + c_2n_y^4 + c_5n_x^2n_y^2),\end{aligned} \quad (61)$$

where $\sigma_m^e = |\mathbf{k}|\tilde{\sigma}_m^e$ and n_x and n_y are the direction cosines of the wave vector, that is, $k_x = |\mathbf{k}|n_x$ and $k_y = |\mathbf{k}|n_y$. We recall that these derivatives should be evaluated for each eigenvalue σ_m^e of the undamped system. For example, we can evaluate derivatives (59) and (60) for the case when $n_y = 0$ and $n_x = 1$. This case corresponds to waves propagating in the direction of the x axis. Such derivatives are easily obtained

and turn out to be independent of the elastic constants, as shown by the following expressions:

$$\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} = -p^{(y/x)}, \qquad \left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} = -1 \quad \text{for } \sigma_m^e = 0;$$
$$\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} = -1, \qquad \left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} = -p^{(x/y)} \quad \text{for } \sigma_m^e \neq 0. \tag{62}$$

Similar results are obtained for the case of waves propagating in the direction of the y axis ($n_x = 0$ and $n_y = 1$):

$$\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} = -1, \qquad \left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} = -p^{(x/y)} \quad \text{for } \sigma_m^e = 0;$$
$$\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} = -p^{(y/x)}, \qquad \left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} = -1 \quad \text{for } \sigma_m^e \neq 0. \tag{63}$$

These particular (limiting) cases make apparent that the classical PML is indeed an unstable medium. Expressions (62) and (63) indicate that the classical PML ($p^{(y/x)} = p^{(x/y)} = 0$) can have an eigenvalue with multiplicity four with zero real part, in addition to the zero eigenvalue observed in (50). In the case of a classical PML for an isotropic medium, these are the only conditions for which the real part of the eigenvalues will be nonnegative. This conclusion can be easily reached when the eigenderivatives (59) and (60) are evaluated for the special case of an M-PML for an isotropic elastic medium, as follows:

$$\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} = -(p^{(y/x)}n_x^2 + n_y^2),$$
$$\left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} = -(n_x^2 + p^{(x/y)}n_y^2) \quad \text{for } \sigma_m^e = 0;$$
$$\left.\frac{d\sigma_m}{dd_x^{(x)}}\right|_{d_x^{(x)}=0} = -(n_x^2 + p^{(y/x)}n_y^2),$$
$$\left.\frac{d\sigma_m}{dd_y^{(y)}}\right|_{d_y^{(y)}=0} = -(p^{(x/y)}n_x^2 + n_y^2) \quad \text{for } \sigma_m^e \neq 0. \tag{64}$$

The preceding derivatives depend only on the direction of propagation and ratios of damping coefficients. In Figure 2 these derivatives have been plotted as a function of the angle of incidence θ (the angle between the vector $\mathbf{k}$ and the x axis, that is, $\cos\theta = n_x$). Figure 2a and b correspond to the eigenderivatives for the classical PML (i.e., $p^{(y/x)} = p^{(x/y)} = 0$), whereas Figure 2c and d show the derivatives for the M-PML with $p^{(y/x)} = p^{(x/y)} = 0.1$. Clearly, if the ratios of damping coefficients are nonzero, the eigenderivatives (64) are negative for all directions of propagation, and therefore for small damping coefficients the real part of the eigenvalues $\{\sigma_m\}$ is negative and consequently the M-PML for isotropic media is asymptotically stable. On the other hand, if the ratios of damping coefficients are zero (classical PML), the derivatives (64) will be zero for waves propagating parallel either to the x axis or to the y axis, as Figure 2a and b shows. Consequently, the classical PML medium can have eigenvalues with zero real part of multiplicity higher than one. Such eigenvalues explain the late-time instability that has been reported in practical applications of the classical PML for isotropic elastic media. Note that because the eigenvalues corresponding to the physical qS and qP modes in the classical PML can indeed have zero real part, the late-time instability is not related to the additional zero eigenvalues introduced in split PML formulations.

Now we turn our attention to classical PML for more general orthotropic media. Those models not only exhibit the instability due to multiple eigenvalues with zero real part mentioned in the preceding discussion (polynomial growth; see equation A.4 in the Appendix), but may also develop instabilities that are associated with eigenvalues having positive real part (exponential growth) and thus become evident considerably earlier during a simulation. We illustrate this second type of instability with a few examples. In particular, we consider the five PML models for orthotropic media investigated by Bécache *et al.* (2003) (denoted by the Roman numerals I, II, III, VI, and V in Bécache *et al.* [2003]) and the models for apatite and zinc reported by Komatitsch and Martin (2007). The properties of the models are listed in Table 1. It is important to remark that the derivatives (59) and (60) depend on the ratio of the elastic coefficients and not on their absolute values.

The eigenderivatives of a classical PML for models I, II, IV, and V as a function of the angle of incidence have been plotted in Figures (3), (4), (5), and (6), respectively. In the cases of these more general orthotropic models, the PML eigenderivatives related to the qP and qS modes take different values. Figures (3)–(6) clearly illustrate that the models I, II, IV, and V exhibit only the instability related to eigenvalues with zero real part of multiplicity higher than one. As it was shown for the isotropic case, any small nonzero value for the ratios $p^{(y/x)}$ and $p^{(x/y)}$ (say 0.1) will make the four media asymptotically stable. On the other hand, the classical PML eigenderivatives for model III can take positive values for several finite ranges of θ, as shown in Figure 7. Therefore, the real parts of the corresponding eigenvalues will take positive values, and the solution will exhibit exponential growth. In the case of the vertical termination strip, Figure 7a indicates that the derivatives related to the nonpropagating mode and the qS mode take positive values for the ranges (0°, 16°) and (45°, 90°), respectively. The values of the derivative corresponding to the nonpropagating mode are relatively closer to zero. However, the positive values of the derivative related to the qS mode are relatively higher, greater than 0.1 for some angles, leading to eigenvalues with a positive real part of higher magnitude (i.e., higher rates of growth), and therefore, the instability associated to that mode will become evident first. The same trend is observed in the

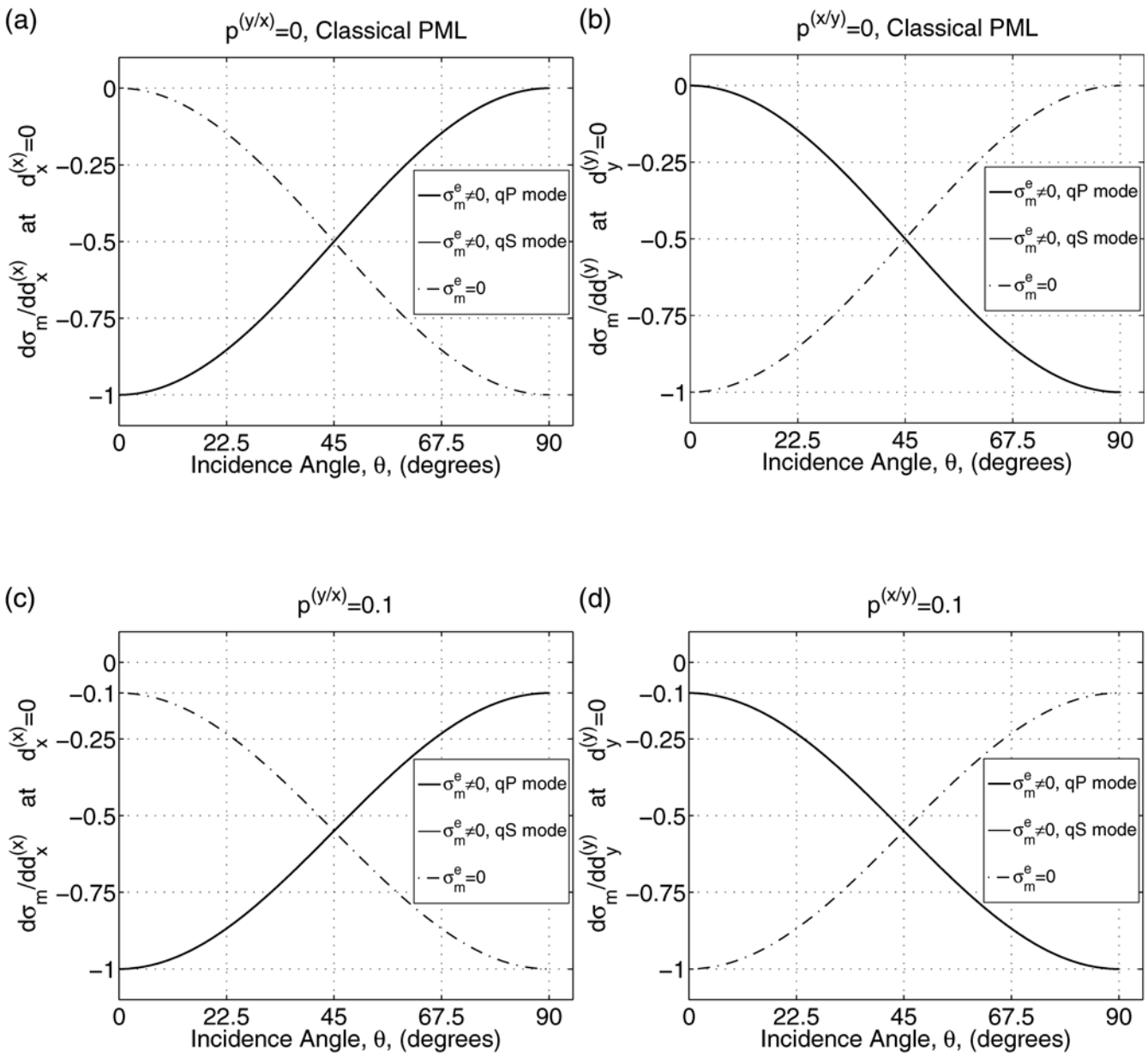


Figure 2. M-PML eigenderivatives for isotropic elastic media. (a) and (b) correspond to the classical PML ($p^{(y/x)} = p^{(x/y)} = 0$). (c) and (d) correspond to the M-PML with $p^{(y/x)} = p^{(x/y)} = 0.1$. In the case of M-PML for isotropic media, the eigenderivatives related to the quasi-*P* (*qP*) and quasi-*S* (*qS*) modes are identical. As it is clearly evident from (c) and (d), the eigenvalues of the M-PML with $p^{(y/x)} = p^{(x/y)} = 0.1$ have a negative real part for all incidence angles, and thus the medium is asymptotically stable.

Table 1

Elastic Coefficients for Selected Orthotropic Models

	Model						
Elastic Constants (N/m^2)	I	II	III	IV	V	Apatite	Zinc
c_{11}	4	20	4	10	30	1.67E+11	1.65E+11
c_{22}	20	20	20	20	6	1.40E+11	6.20E+10
c_{33}	2	2	2	6	1.5	6.63E+10	3.96E+10
c_{12}	3.80	3.8	7.5	2	9.9	6.60E+10	5.00E+10

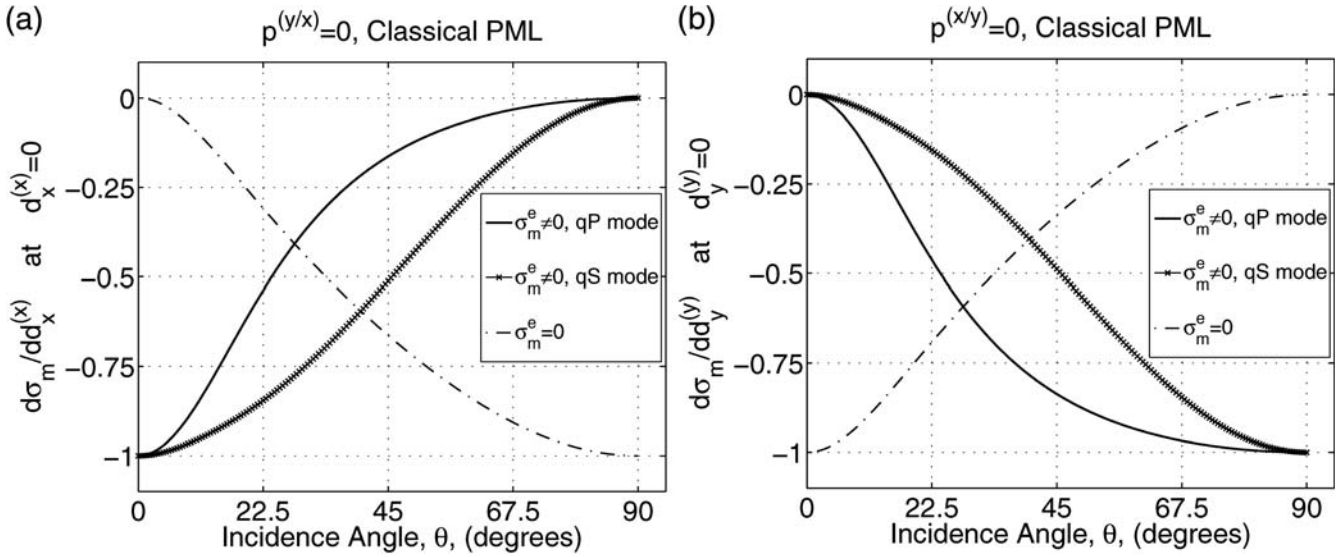


Figure 3. Classical PML eigenderivatives for an orthotropic elastic medium (model I). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical strip. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal strip. Both vertical and horizontal termination strips exhibit only the instability due to eigenvalues with zero real part of multiplicity greater than one.

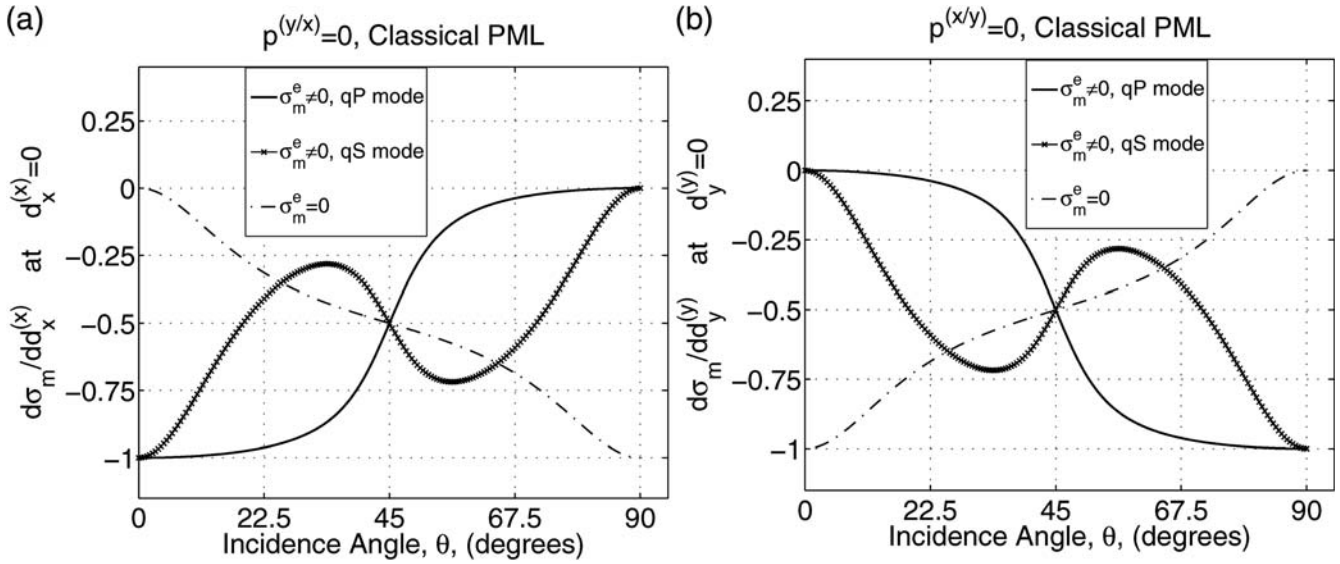


Figure 4. Classical PML eigenderivatives for an orthotropic elastic medium (model II). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip. Both vertical and horizontal termination strips exhibit only the instability due to eigenvalues with zero real part of multiplicity greater than one.

eigenderivatives for the horizontal termination strip, as shown in Figure 7b. The derivatives associated with the zero eigenvalue and the qS mode take positive values for the ranges (58°, 90°) and (0°, 29°), respectively. The derivative associated to the zero eigenvalue again remains close to zero whereas the one related to the qS mode can take values of about 0.34 for some angles. Thus, for the horizontal termination strip, the exponential growth of the qS mode will be observed first.

In order to design a stable M-PML for the orthotropic model III, we evaluate the eigenderivatives with ratios $p^{(y/x)}$ and $p^{(x/y)}$ different from zero. As Figure 8a and b illustrate, using ratios $p^{(y/x)}$ and $p^{(x/y)}$ both equal to 0.15 stabilizes the vertical termination strip, but not the horizontal

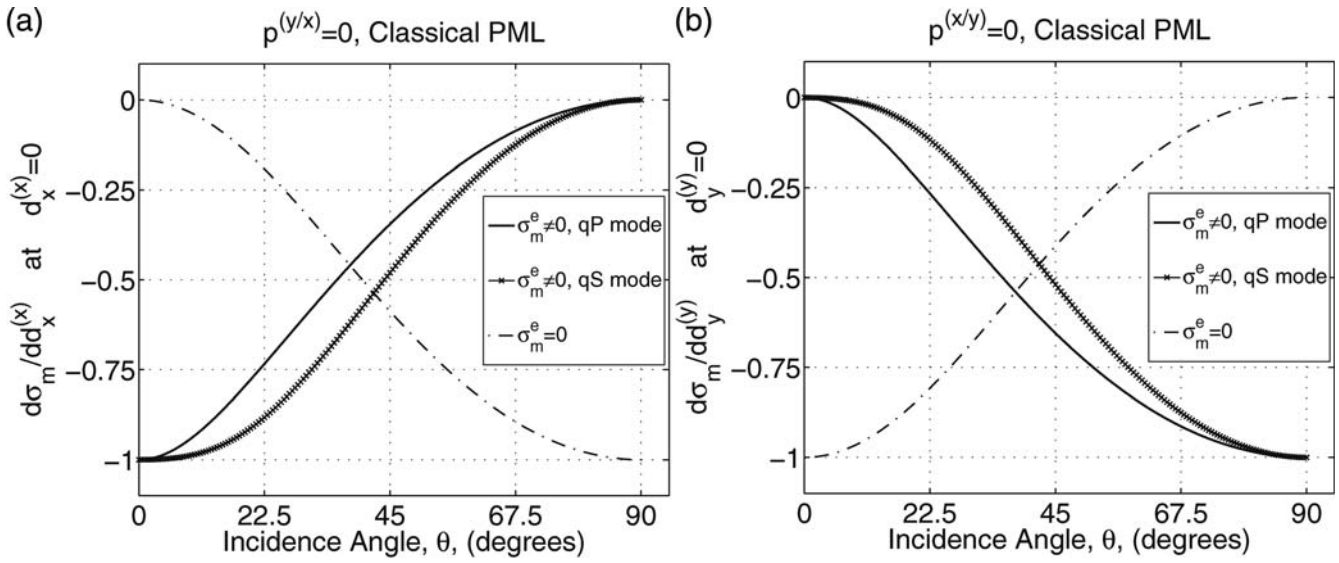


Figure 5. Classical PML eigenderivatives for an orthotropic elastic medium (model IV). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip. Both vertical and horizontal termination strips exhibit only the instability due to eigenvalues with zero real part of multiplicity greater than one.

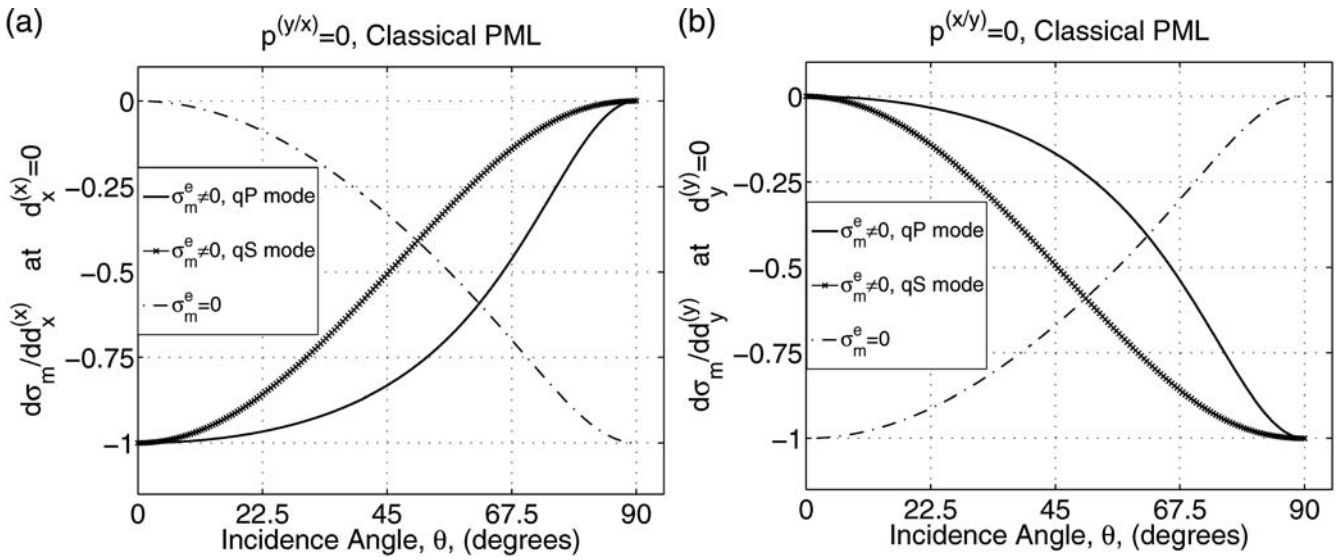


Figure 6. Classical PML eigenderivatives for an orthotropic elastic medium (model V). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip. Both vertical and horizontal termination strips exhibit only the instability due to eigenvalues with zero real part of multiplicity greater than one.

one. For the latter, the maximum value of the eigenderivative associated with the qS mode still takes positive values for the range of (9°, 26°). Thus, the eigenderivative for the horizontal termination strip is recalculated with $p^{(x/y)} = 0.3$, and the results are shown in Figure 8c. This time the derivatives associated with all modes are negative for all values of the incidence angle, and therefore, the horizontal M-PML is asymptotically stable.

Figure 9a and b and Figure 10a and b show that classical PML models for apatite and zinc also exhibit exponential growth. For both cases, the positive eigenvalues are related to the qS mode and are present in both the horizontal and

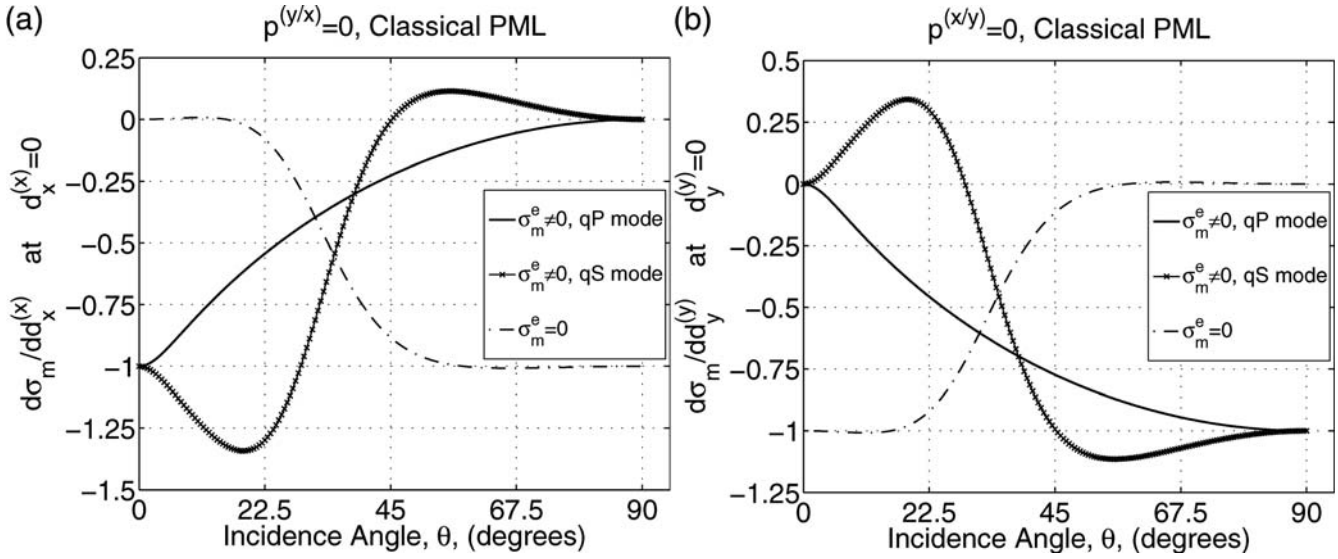


Figure 7. Classical PML eigenderivatives for an orthotropic elastic medium (model III). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip. Both horizontal and vertical termination strips exhibit the instability related to the polynomial growth as well as the exponential growth instability.

vertical termination strips. As it can be observed in Figure 9c and d, a value of 0.1 for the ratios $p^{(y/x)}$ and $p^{(x/y)}$ is sufficient to stabilize the model for apatite. In the case of the M-PML model for zinc, $p^{(x/y)} = 0.1$ stabilizes the horizontal termination strip, as shown in Figure 10d. For the vertical termination strip to become asymptotically stable, however, $p^{(y/x)} = 0.15$ seems to be the proper value, as Figure 10c indicates.

We have shown that with the simple eigenvalue sensitivity analysis one can design robust M-PML terminations, that is, media that are asymptotically stable. It should be pointed out, however, that prediction of the sign of the real part of the eigenvalues based on the sign of the first eigenderivative is valid only if the damping coefficients are small (close to zero). If the damping coefficients are large enough to allow the eigenvalues to oscillate and change sign as they move away from the values that they have for zero damping, then the analysis presented in the preceding discussion based on the first derivatives is not valid any more. Such cases may be encountered when the damping profiles of the M-PML are specified to grow very fast, and much higher values of the ratios $p^{(y/x)}$ and $p^{(x/y)}$ would be required to stabilize such media. However, because very high damping ratio amplitudes increase the reflections due to the numerical dispersion that exists at the discrete level of the PML terminations (e.g., Collino and Tsogka, 2001), using small damping ratios, and therefore, reasonably thicker M-PMLs for stabilization, seems to be a preferable option.

Numerical Experiments

In order to validate the stability analysis presented in the previous section and test its effectiveness, we consider three numerical experiments. In the first test we study the stability of the M-PML for a two-dimensional isotropic elastic medium with a free surface. For comparison, results of the test using the C-PML are also presented. In the other two tests, we simulate waves propagating in two-dimensional square models representing the orthotropic elastic medium III and zinc. The equations in the physical domain and M-PML terminations were solved in variational form with the spectral element method (SEM). For implementation of the split-field M-PML in the SEM, the staggered velocity-stress time scheme proposed by Festa and Vilotte (2005) was adopted. For all simulations, we selected a quadratic damping profile of the form

$$d_x^{(x)} = d_0\left(\frac{x}{H}\right)^2, \qquad d_y^{(y)} = d_0\left(\frac{y}{H}\right)^2, \tag{65}$$

where H is the thickness of the absorbing termination strip and the parameter d_0 is the maximum value of the damping profile in the strip. Furthermore, the source time variation is given by a Ricker wavelet having the following waveform:

$$R(t) = [2\pi^2 f_0^2 (T - t_0)^2 - 1]\exp[-\pi^2 f_0^2 (T - t_0)^2], \tag{66}$$

where f_0 is the dominant frequency of the signal and t_0 is its onset time.

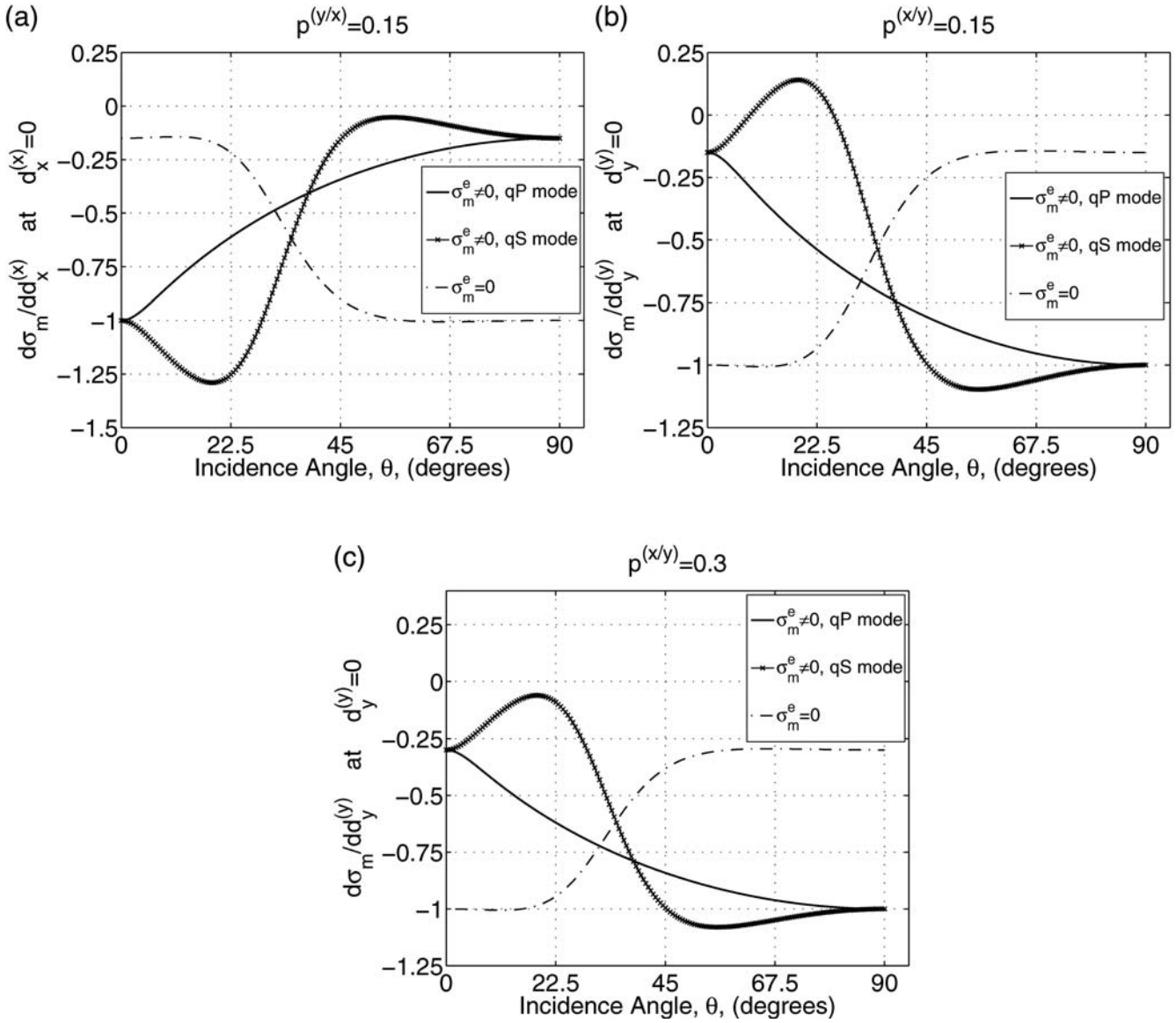


Figure 8. M-PML eigenderivatives for an orthotropic elastic medium (model III). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip with $p^{(y/x)} = 0.15$. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip with $p^{(x/y)} = 0.15$. (c) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip with $p^{(x/y)} = 0.30$.

Finally, in order to illustrate the efficiency of the three different PML methods, the evolution of the total energy (stored in the physical domain) during the simulations is investigated. Thus, the total energy at each time instant is computed according to the following expression:

$$E = \frac{1}{2}\int_{\Omega} \rho \|\mathbf{v}\|^2 \, d\Omega + \frac{1}{2}\int_{\Omega} T_{ij}\varepsilon_{ij} \, d\Omega, \qquad (67)$$

where Ω is the volume corresponding to the physical domain, $\|\mathbf{v}\|$ is magnitude of the velocity vector, and ε_{ij} are the components of the strain tensor $\mathbf{E}$.

Isotropic Medium

An important application of PML media in computational seismology is the simulation of propagation of surface waves in geological basins, which are modeled as isotropic media. If the simulations are performed for long time durations, the multiple-zero instability inherent in the classical PML might arise and pollute the solution in the physical domain. It has been reported (Festa *et al.*, 2005) that implementation of the convolutional (or generalized) PML eliminates the instability. Here, we show that the M-PML is also an efficient alternative solution. We know that the equations of elastodynamics also admit inhomogeneous plane-wave

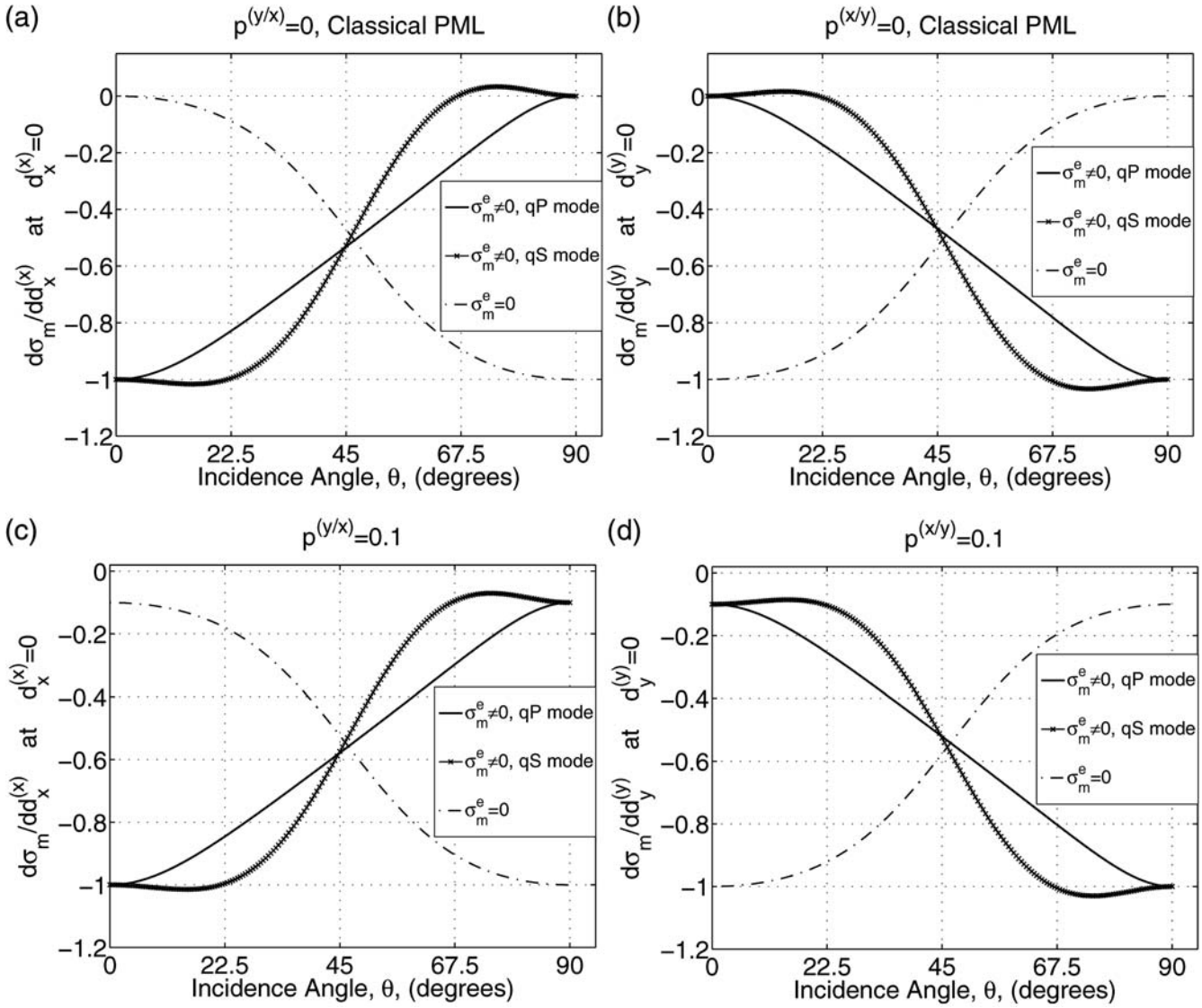


Figure 9. M-PML eigenderivatives for an orthotropic elastic medium (model for apatite). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip with $p^{(y/x)} = 0$. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip with $p^{(x/y)} = 0$. (c) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip with $p^{(y/x)} = 0.1$. (d) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip with $p^{(x/y)} = 0.1$. The classical PML model for both vertical and horizontal termination strips exhibit the instability related to the polynomial growth as well as the exponential growth instability.

solutions. For such type of waves at least one component of the wave vector is imaginary (see, e.g., Aki and Richards, 2002, for a comprehensive discussion of inhomogeneous waves). In this case it is the term k_x (and/or k_y, and/or k_z) that introduces the spatial attenuation. It follows that if the M-PML medium is asymptotically stable, any combination of inhomogeneous waves inside the medium will be also damped out. Thus, it is expected that surface waves, which consist of a superposition of inhomogeneous waves (see, e.g., Aki and Richards, 2002), will be damped out when they enter in the M-PML termination. For demonstration purposes we consider a problem similar to the one studied by Festa *et al.* (2005). Specifically, we consider the two-dimensional problem of a line explosive source embedded in a homogeneous isotropic elastic half-space. Thus, the physical domain in this case is bounded at the top by a free surface and on the other three sides by absorbing layers. The free surface condition was also imposed on the top edges of the vertical termination strips, whereas Dirichlet boundary conditions $\mathbf{v} = \mathbf{0}$ were specified at all other external boundaries of the three termination strips. The explosive source is located close to the left absorbing strip. The dimensions and properties of

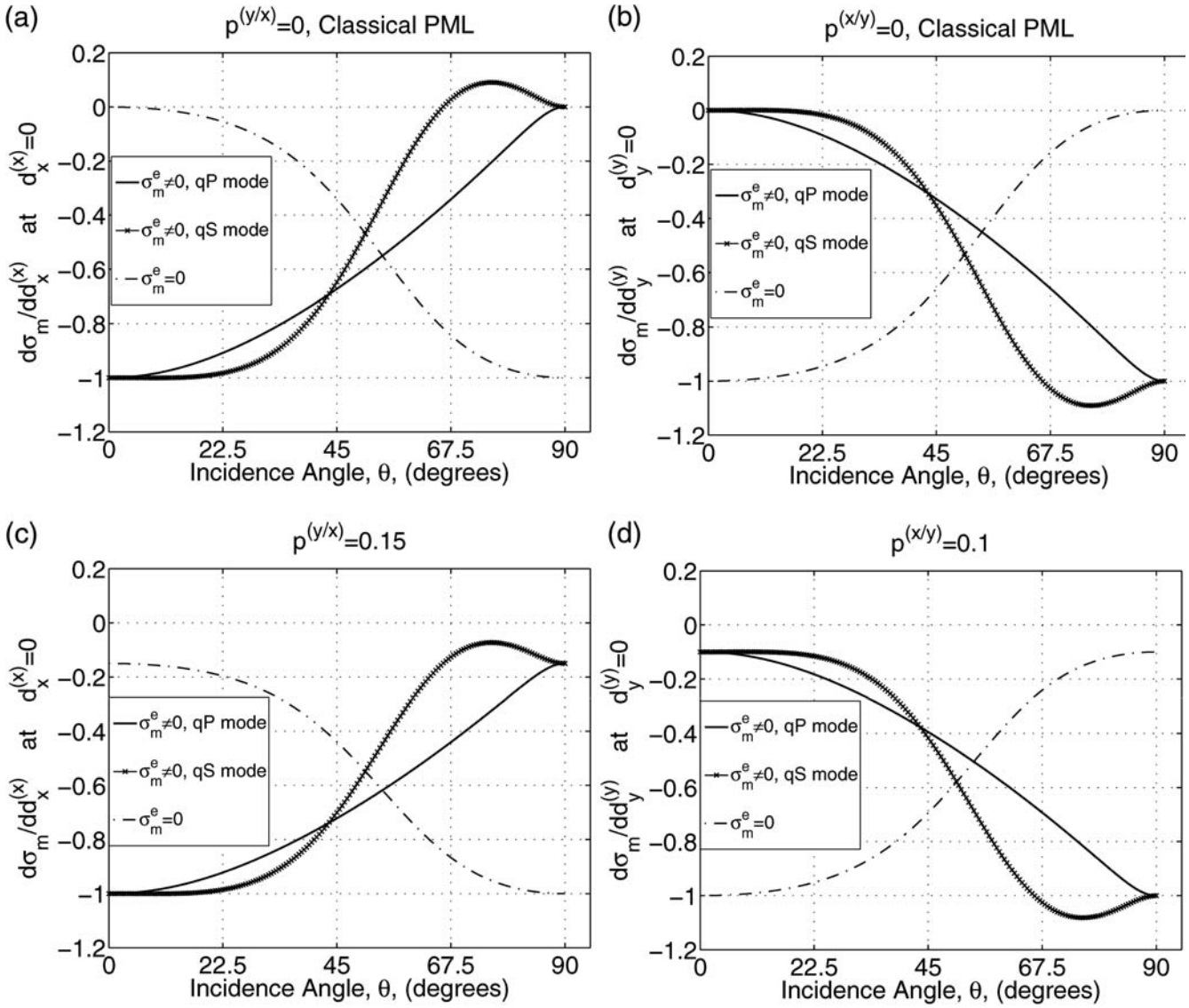


Figure 10. M-PML eigenderivatives for an orthotropic elastic medium (model for zinc). (a) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip with $p^{(y/x)} = 0$. (b) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip with $p^{(x/y)} = 0.1$. (c) Eigenderivatives with respect to $d_x^{(x)}$ evaluated at $d_x^{(x)} = 0$, for a vertical termination strip with $p^{(y/x)} = 0.15$. (d) Eigenderivatives with respect to $d_y^{(y)}$ evaluated at $d_y^{(y)} = 0$, for a horizontal termination strip with $p^{(x/y)} = 0.1$. For both vertical and horizontal termination strips the classical PML model exhibits the instability related to the polynomial growth as well as the exponential growth instability.

the media and discretization parameters are listed in Table 2. The damping profiles are given by expressions (65) with a maximum value d_0 given by the following expression (Festa and Vilotte, 2005):

$$d_0 = A\frac{v_p}{H} \tag{68}$$

with $A = 10$. We performed the simulations with classical PML, M-PML with $p^{(y/x)} = p^{(x/y)} = 0.1$, and C-PML terminations. For the latter we adopted the same damping profile (65) and a cutoff frequency of 1.5 Hz.

In Figure 11 snapshots of the results are displayed for different instants of time. The interfaces between the physical domain and termination strips are represented by the solid lines. It can be observed that the body and surface waves are well absorbed by the three termination strips. At $t = 8.5$ sec, the late-time instability of the classical PML is visible. If the simulation is performed for longer times, the instability grows and spreads into the physical domain. No instabilities are identified in the C-PML and M-PML terminations.

Table 2

Properties and Discretization Parameters for Simulation in Isotropic Medium

Parameter	Value
Physical domain dimensions	
Length	11 km
Width	1 km
Physical domain properties	
Density	2.7E+12 kg/km^3
S–wave velocity	1.87 km/ sec
P–wave velocity	3.2 km/ sec
Source location	
From left boundary	0.25 km
From bottom boundary	3.75 km
Ricker wavelet parameters	
Dominant frequency	2.5 Hz
Onset time	0.4 sec
Discretization parameters	
Polynomial degree	8
Element side	0.1 km
Elements along PML width	10
Timestep	0.00047 sec
Total duration	8.5 sec

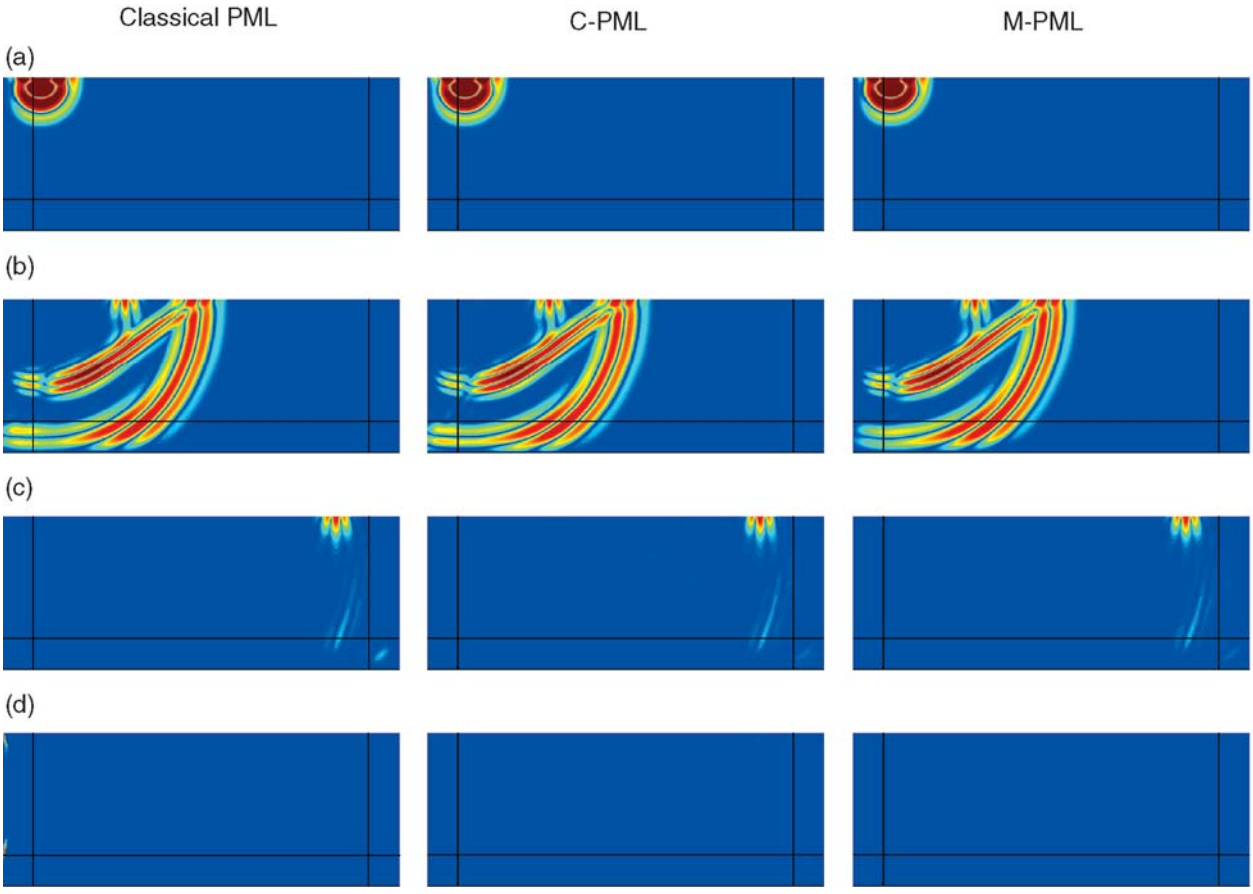


Figure 11. Snapshots of propagation of the velocity magnitude in an isotropic elastic medium (width 4 km) at (a) $t = 0.5$ sec, (b) $t = 2$ sec, (c) $t = 6$ sec, and (d) $t = 8.5$ sec. The panels on the right, center, and left correspond to classical PML, C-PML, and M-PML ($\{p^{(y/x)} = p^{(x/y)} = 0.1\}$) terminations, respectively. As shown in (d), at $t = 8.5$ sec the late-time instability of the classical PML can be observed on the left vertical termination strip. No instabilities are identified in the C-PML and M-PML terminations.

It has been argued (Festa *et al.*, 2005) that the instability observed in the classical PML, for simulations that involve surface waves, is the result of the strong interaction of the surface wave with the termination lying at the bottom of the physical domain. For this reason we repeated the same simulation, but this time the depth of the physical domain was only 1 km so as to impose strong interaction between the generated surface wave and the bottom termination. The results are displayed in Figure 12. It is evident from this figure that the strong interaction does not change the behavior of the PML models. For both values of depth of the physical domain (i.e., 4 and 1 km), the instability manifests itself for the first time in the left classical PML termination, while no instabilities are observed in any C-PML or M-PML termination.

Figure 13 displays the energy decay in the physical domain for the three PML models under investigation for the two different depths that were considered (4 and 1 km). As Figure 13b and d indicaes, after the wave field has left the physical domain (which occurs at about 6.5 sec) the lowest decay rate is observed in the case of the C-PML. The lower decay rate is attributed to wave energy, with frequencies lower than the specified cutoff frequency, which is not absorbed by the C-PML. The absorption in the C-PML could be improved by reducing this cutoff frequency. There is no such problem when M-PML terminations are implemented. Both types of termination strips appear to be stable for the isotropic medium under investigation. However, let us mention that although simple unsplit formulations for C-PML can be used for isotropic media (Drossaert and Giannopolous, 2007; Komatitsch and Martin, 2007), the M-PML has the advantage of being nonconvolutional.

Orthotropic Medium-Model III

The second experiment is related to the classical PML for the orthotropic model III. We perform simulations to illustrate the instabilities and to confirm that a stable M-PML termination for this model can be constructed. As far as we know, a stable PML for this particular orthotropic model has not been designed in the past. A square elastic domain is considered, surrounded by absorbing layers on its four sides. The M-PMLs are in turn terminated with Dirichlet boundary conditions $\mathbf{v} = \mathbf{0}$. An explosive source is located near the top left corner of the domain. The maximum value d_0 of the damping profiles used for the simulations is 25. More details on the properties of the medium and discretization are given in Table 3.

Snapshots of the results of the numerical experiments using the classical PML and two different M-PML terminations are given in Figure 14. As stated in the previous section, the response of the classical PML for the orthotropic model III exhibits the exponential growth. This instability becomes visible at $t = 3.5$ sec, early in the simulation, starting in the horizontal (top) termination strip and then appearing at other locations inside the classical PML. By the time $t = 6.5$ sec the spurious signals have already entered the physical domain and the simulation using the classical PML is stopped. The snapshots of the simulation using an M-PML with $p^{(y/x)} = p^{(x/y)} = 0.15$ (center column of Fig. 14) show that this medium exhibits instabilities as well. The difference

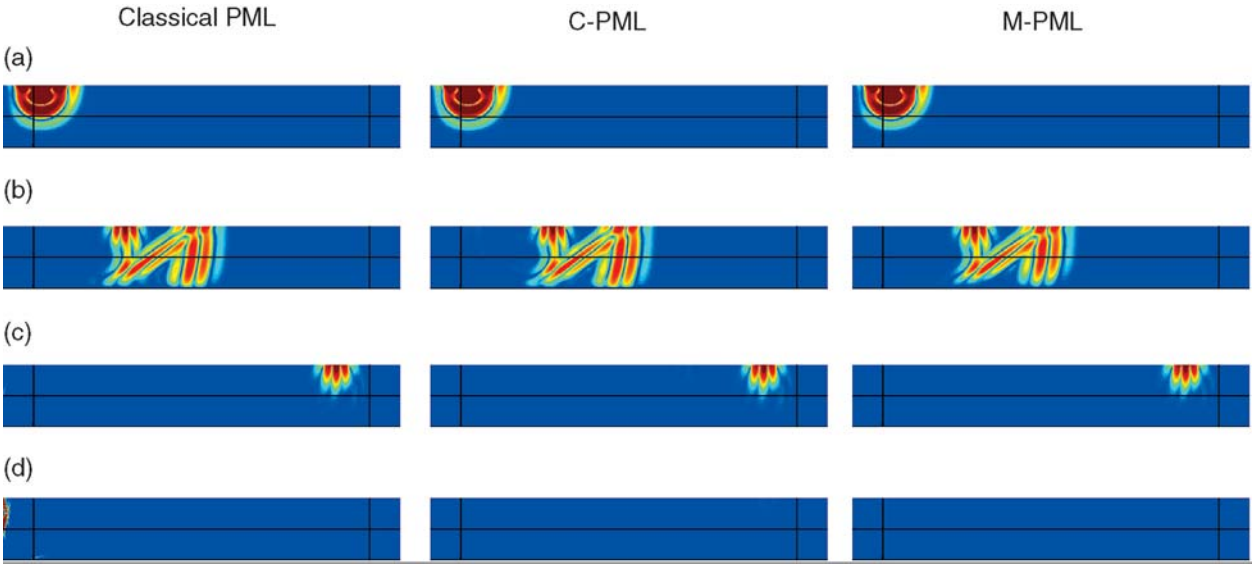


Figure 12. Snapshots of propagation of the velocity magnitude in an isotropic elastic medium (width 1 km) at (a) $t = 0.5$ sec, (b) $t = 2$ sec, (c) $t = 6$ sec, and (d) $t = 8.5$ sec. The panels on the right, center, and left correspond to classical PML, C-PML, and M-PML ($\{p^{(y/x)} = p^{(x/y)} = 0.1\}$) terminations, respectively. As shown in (d), at $t = 8.5$ sec the late-time instability of the classical PML can be observed on the left vertical termination strip. No instabilities are identified in the C-PML and M-PML terminations.

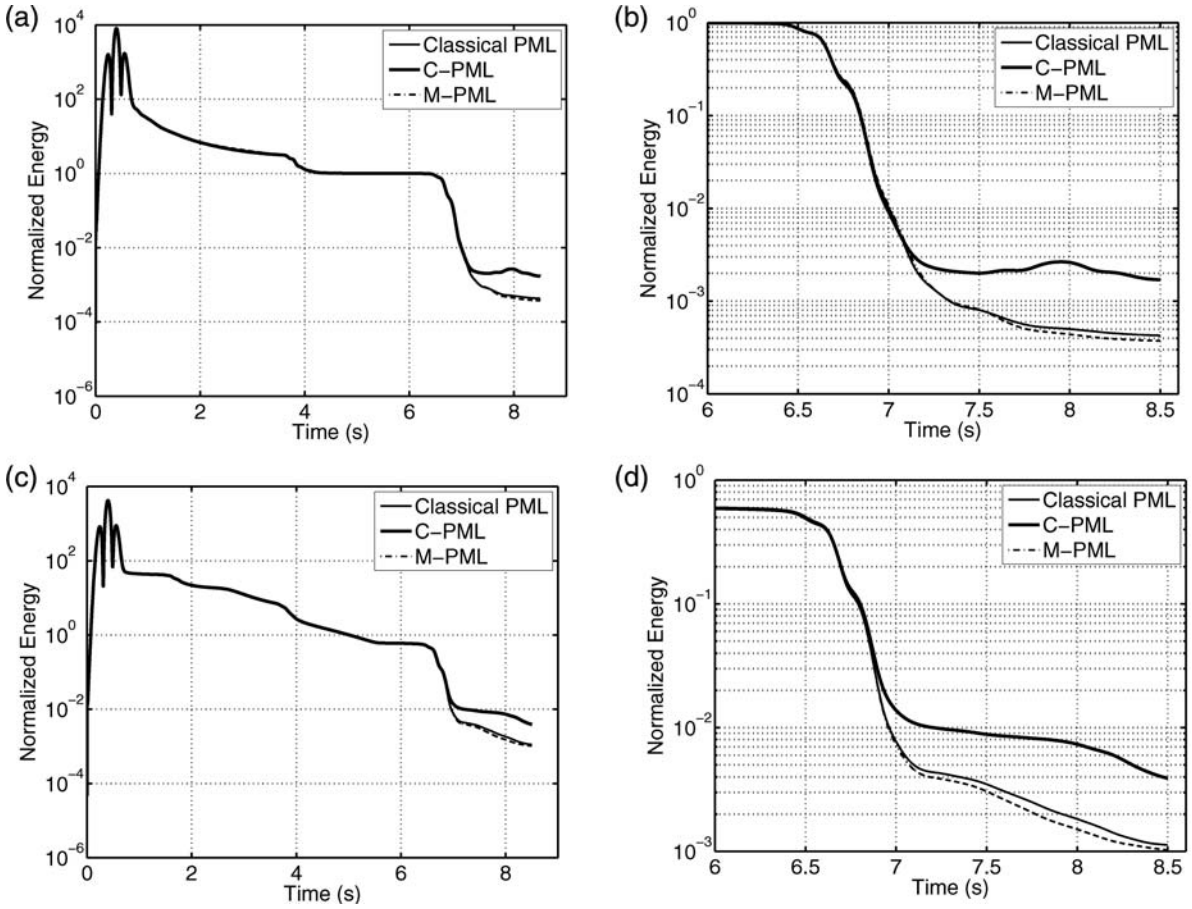


Figure 13. Energy decay in the physical domain for the isotropic medium. The energy is normalized with respect to the value that it attains at time $t = 5$ sec. (a) Energy decay in the physical domain of depth equal to 1 km. (b) Energy decay in the physical domain of depth equal to 1 km over the time interval 6–8.5 sec. (c) Energy decay in the physical domain of depth equal to 4 km. (d) Energy decay in the physical domain of depth equal to 4 km over the time interval 6–8.5 sec.

Table 3

Properties and Discretization Parameters for Simulation in Model III

Parameter	Value
Physical domain dimensions	
Length	8 m
Width	8 m
Physical domain properties	
Density	1 kg/m^3
c_{11}	4 N/m^2
c_{22}	20 N/m^2
c_{33}	2 N/m^2
c_{12}	7.5 N/m^2
Source location	
From left boundary	0.15 m
From bottom boundary	7.85 m
Ricker wavelet parameters	
Dominant frequency	0.9 Hz
Onset time	1.1 sec
Discretization parameters	
Polynomial degree	5
Element side	0.25 m
Elements along PML width	7
Timestep	0.0033 sec
Total duration	40 sec

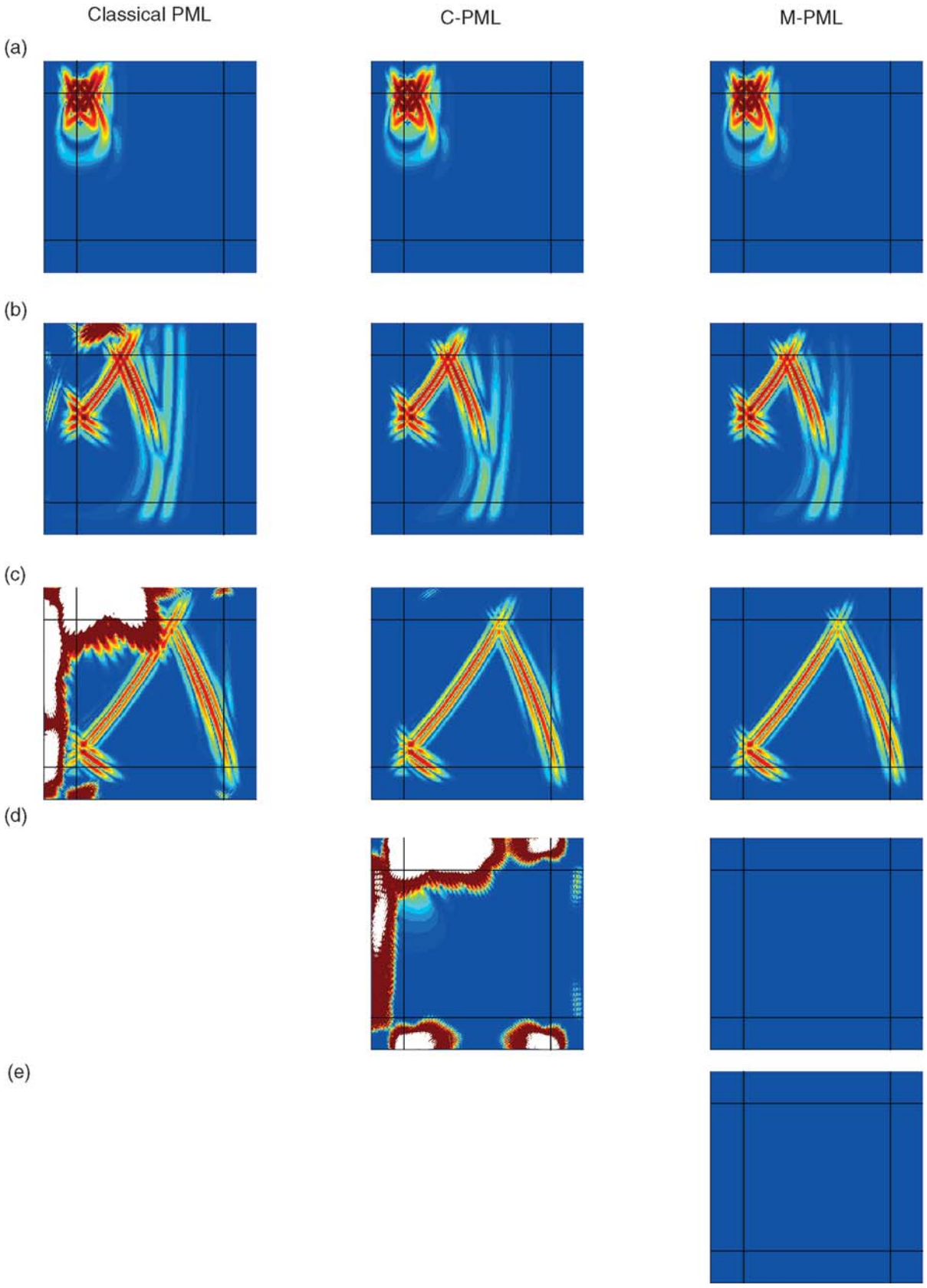


Figure 14. Snapshots of propagation of the velocity magnitude in an orthotropic elastic medium, model III, at (a) $t = 1.5$ sec, (b) $t = 3.5$ sec, (c) $t = 6.5$ sec, (d) $t = 18$ sec, and (e) $t = 40$ sec. The panels on the left correspond to classical PML terminations. The panels in the center and on the right correspond to M-PML terminations with $\{p^{(y/x)} = p^{(x/y)} = 0.15\}$ and $\{p^{(y/x)} = 0.30, p^{(x/y)} = 0.25\}$, respectively. No instability is observed for the M-PML terminations on the right-hand panels.

with the classical PML is that the instabilities in this M-PML become visible at later times. Thus, the effect of increasing the ratios of damping profiles from 0 to 0.15 is the reduction of the instabilities rate of growth. The instability in the top horizontal termination strip appears at $t = 6.5$ sec. At $t = 18$ sec the instabilities have appeared at other locations inside the M-PML, propagated inside the physical domain, and eventually the simulation is stopped. It should be noted that according to the eigenvalue sensitivity analysis for this medium the exponential growth was expected only in the horizontal termination strip (see Fig. 8b) and not on the vertical termination strip (see Fig. 8a). However, as it is evident in Figure 14d, instabilities appear in both the horizontal as well as in the vertical termination strip. This is an example of an M-PML for which the analysis based on the first eigenderivatives fails to predict the optimum stabilizing ratio $p^{(x/y)}$. The reason for this failure is the high value ($d_0 = 25$) of the damping coefficient d_0 that was used. We recall that the predictive scheme that we developed in the previous sections based on the sign of the first eigenderivative is not valid for large values of the damping coefficient. However, the information given by this predictive scheme may still be a good starting point toward stabilizing the medium. Specifically, by increasing the values of the damping ratios $p^{(y/x)}$ and $p^{(x/y)}$ while keeping the value of d_0 the same (i.e., equal to 25), we managed to stabilize the medium. The right-hand column of Figure 14 displays the results of the simulations using an M-PML with $p^{(y/x)} = 0.30$ and $p^{(x/y)} = 0.25$. The simulation was performed up to a time of 40 sec, and no instability was observed in any of the absorbing layers. An alternative way to stabilize the medium using a lower value for $p^{(x/y)}$ is the reduction of the parameter d_0 and the increase of the width of the M-PML layer, if needed. Finally the energy decay in the physical domain with the stable M-PML terminations is displayed in Figure 15.

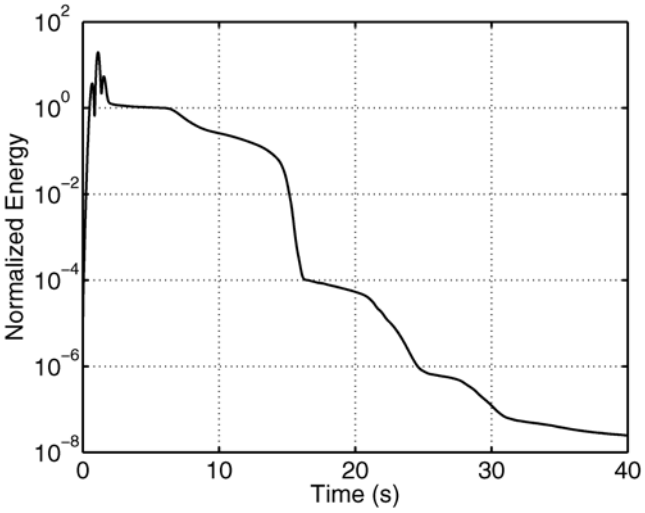


Figure 15. Energy decay in the physical domain for model III. The energy is normalized with respect to the value that it attains at time $t = 6$ sec.

Orthotropic Media: Zinc

Komatitsch and Martin (2007) reported the classical PML and C-PML are unstable for numerical simulations involving models of common crystals like zinc. In this numerical experiment we illustrate again that it is possible to construct stable M-PMLs for such orthotropic media. The configuration of the test for zinc is similar to that presented by Komatitsch and Martin (2007). The physical domain is a square surrounded by M-PMLs on its four sides. The M-PMLs are again terminated with Dirichlet boundary conditions $\mathbf{v} = \mathbf{0}$. The source is a concentrated vertical force acting at the center of the physical domain. The maximum value d_0 of the damping profile adopted for this test is 0.47. Table 4 provides more details on the properties of the medium and discretization parameters.

Snapshots of the results of the numerical experiments at different times are displayed in Figure 16. It can be observed that both the qP and qS waves are well absorbed by both layers, namely, the classical PML and the M-PML with $p^{(y/x)} = p^{(x/y)} = 0.15$. At time instant $t = 400$ μsec (Fig. 16d, left-hand column), the exponential growth is visible in both the vertical and the horizontal termination strips. On the other hand, no instabilities are detected in the snapshots for the M-PML (Fig. 16, right-hand column). Because of the small value of d_0 specified for the M-PMLs in this test, the results are those anticipated by the eigenvalue sensitivity

Table 4
Properties and Discretization Parameters for Simulation in Zinc

Parameter	Value
Physical domain dimensions	
Length	25 cm
Width	25 cm
Physical domain properties	
Density	7100 kg/m^3
c_{11}	1.65E+11 N/m^2
c_{22}	6.20E+10 N/m^2
c_{33}	3.96E+10 N/m^2
c_{12}	5.00E+10 N/m^2
Source location	
From left boundary	12.5 cm
From bottom boundary	12.5 cm
Ricker wavelet parameters	
Dominant frequency	170 kHz
Onset time	5.88 μsec
Discretization parameters	
Polynomial degree	5
Element side	0.625 cm
Elements along PML width	10
Timestep	0.04 μsec
Total duration	400 μsec

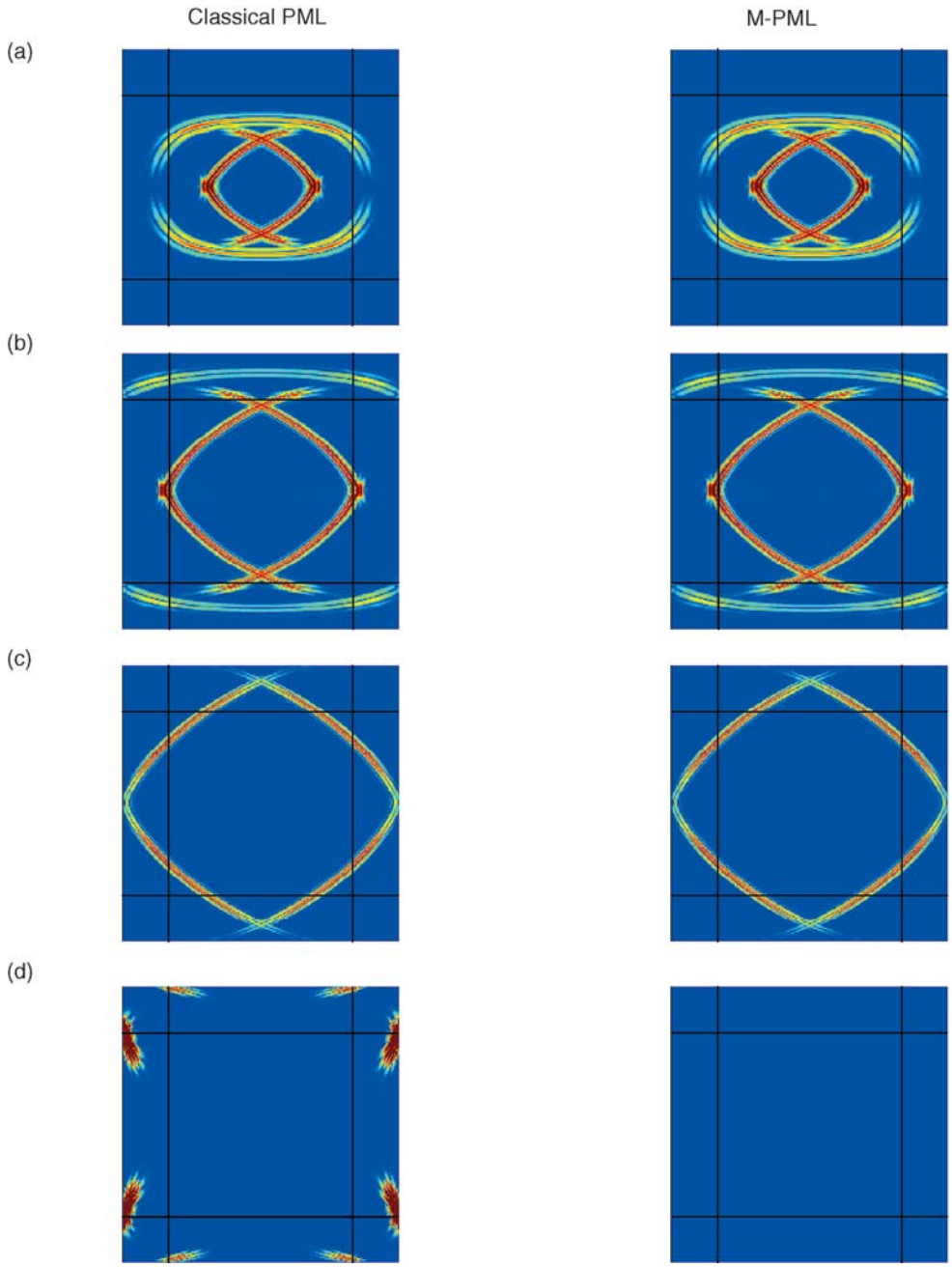


Figure 16. Snapshots of propagation of the velocity magnitude in orthotropic elastic medium, zinc model, at (a) $t = 35\ \mu$sec, (b) $t = 60\ \mu$sec, (c) $t = 85\ \mu$sec, and (d) $t = 400\ \mu$sec. The right-hand and left-hand panels correspond to classical PML and M-PML (with $p^{(y/x)} = p^{(x/y)} = 0.15$) terminations, respectively. No instability is observed in the M-PML terminations.

analysis presented in the previous section (see Fig. 10a and b). Finally the energy decay in the physical domain with the M-PML terminations is displayed in Figure 17.

Conclusions

A nonconvolutional, split-field perfectly matched layer, referred to as the multiaxial perfectly matched layer, has been presented. The new formulation is the result of the generalization of the classical PML to a medium in which damping profiles are specified in more than one direction. Under the hypothesis of small damping coefficients, we proposed an eigenvalue sensitivity analysis based on first derivatives to provide effective criteria to study the stability of the M-PML. With such a method we demonstrated that the stability of the M-PML can be related to the ratios of the specified damping profiles, and thus we provided a general procedure to construct robust, stable M-PML models for anisotropic media. It has also been demonstrated that for any anisotropic medium the classical PML exhibits

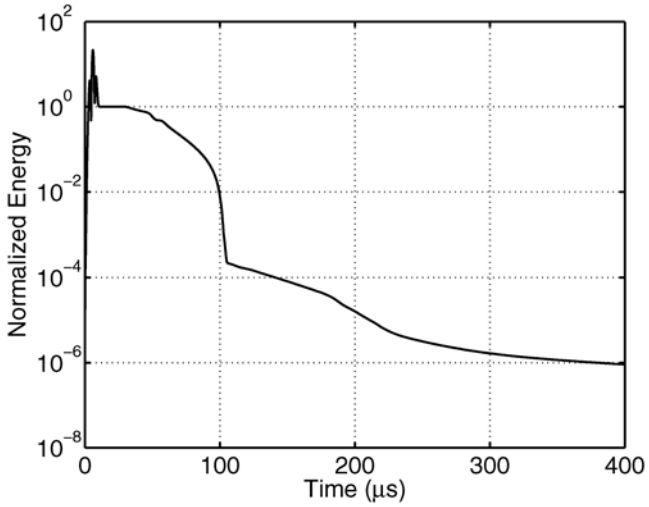


Figure 17. Energy decay in the physical domain for zinc. The energy is normalized with respect to the value that it attains at time $t = 12$ μsec.

instabilities related to eigenvalues with zero real part of multiplicity higher than one. Furthermore, we showed that exponential growth due to eigenvalues with positive real part can be present in the classical PML for some orthotropic media. Stable M-PML models were then constructed, and optimum ratios of damping profiles were proposed. Through numerical experiments involving isotropic and orthotropic media, it was verified that stable absorbing boundaries can be constructed with the proposed formulation. The resulting split-field M-PML models for anisotropic elastodynamics remain stable for all frequencies and directions of propagation. The generality of the proposed method renders it suitable to study the stability of PML models for wave propagation problems in other fields such as acoustics and electromagnetism. Implementation of the proposed M-PML model has been demonstrated for two-dimensional problems, and the predictive scheme for investigating its stability has been developed for the same kind of problems. Extension of this scheme to three-dimensional problems is straightforward and will be reported in a future publication.

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Appendix

Stability Definitions

For completeness, in this appendix we provide stability definitions. The definitions presented herein and more detailed accounts on the subject can be found in Sastry (1999) or other works on stability of dynamical systems.

Stability in the Sense of Lyapunov

The equilibrium point $\mathbf{U}(t) = \mathbf{0}$ is called a stable equilibrium point of (37) if for all $t_0 \geq 0$ and $\varepsilon > 0$ there exists $\delta(t_0, \varepsilon)$ such that

$$|\mathbf{U}_0| < \delta(t_0, \varepsilon) \Rightarrow |\mathbf{U}(t)| < \varepsilon \quad \forall\ t \geq t_0, \tag{A.1}$$

where $\mathbf{U}_0$ is the vector describing the initial condition at $t = t_0$. This means that $\mathbf{U}(t) = \mathbf{0}$ is a stable equilibrium point if the solutions $\mathbf{U}(t)$ of (37) remain close to the origin when the initial condition $\mathbf{U}_0$ is close enough to the origin. This notion of stability is weak in the sense that it does not require trajectories starting close to the origin to asymptotically converge to the origin. Such a feature is included in the definition of asymptotic stability that follows.

Asymptotic Stability

The equilibrium point $\mathbf{U} = \mathbf{0}$ is an asymptotically stable equilibrium point of (37) if

1. $\mathbf{U} = \mathbf{0}$ is a stable equilibrium point of (37),
2. $\mathbf{U} = \mathbf{0}$ is attractive, that is, for all $t_0 \geq 0$ there exists a $\delta(t_0)$ such that

$$|\mathbf{U}_0| < \delta \Rightarrow \lim_{t \to \infty} |\mathbf{U}(t)| = 0. \tag{A.2}$$

An asymptotically stable equilibrium point is called a sink. For linear systems, asymptotic stability of the equilibrium at the origin is equivalent to the stability of any other solution. Therefore, the stability properties of the solutions may be attributed to the system itself. If an equilibrium point is stable but not asymptotically stable, it is called marginally stable.

Solutions of Autonomous Systems

If the eigenvalues $\{\sigma_i\}$ of $\mathbf{A}$ in system (37) are distinct with corresponding eigenvectors $\{\mathbf{V}_i\}$, $i = 1, \ldots, 10$, then system (37) has 10 independent solutions of the form

$$\mathbf{V}_i \exp(\sigma_i t). \tag{A.3}$$

If not all eigenvalues are distinct, say eigenvalue σ_i has multiplicity $\ell > 1$, then σ_i generates ℓ independent solutions of the form

$$\mathbf{P}_0 \exp(\sigma_i t), \mathbf{P}_1(t) \exp(\sigma_i t), \mathbf{P}_2(t) \exp(\sigma_i t), \ldots, \mathbf{P}_{\ell-1}(t) \exp(\sigma_i t), \tag{A.4}$$

where $\mathbf{P}_k(t)$, $k = 1, \ldots, \ell - 1$, are vectors whose components are polynomials of degree k or smaller. When studying the stability of solution $\mathbf{U} = \mathbf{0}$, it is clear from (A.4) that it is determined by the real part of the eigenvalues $\{\sigma_i\}$. The following theorem provides necessary and sufficient conditions for the asymptotic stability of a linear system.

Theorem.—System (37) is asymptotically stable if and only if

$$\mathrm{Re}\{\sigma_i\} < 0 \quad \forall \ i, \tag{A.5}$$

where $\{\sigma_i\}$ is the set of eigenvalues of the matrix $\mathbf{A}$. Therefore, to demonstrate that an autonomous system is asymptotically stable it suffices to demonstrate that the real part of its eigenvalues is strictly negative.

There is a strong form of stability that demands an exponential rate of convergence. It is referred to as Exponential Stability and is defined as follows.

Exponential Stability

The equilibrium point $\mathbf{U} = \mathbf{0}$ is an exponentially stable equilibrium point of (37) if there are M and $r < 0$ such that

$$|\mathbf{U}(t)| \leq M \exp[r(t - t_0)]|\mathbf{U}_0| \tag{A.6}$$

for all $\mathbf{U}_0$ and $0 \leq t_0 \leq t$. The constant parameter r is called the rate of convergence. Systems that are exponentially stable are desirable for applications because they are robust to perturbations. In contrast, the stability of a solution that is not exponentially stable can be destroyed by arbitrary small perturbations, such as numerical errors (Hinrichsen and Pritchard, 2005). Fortunately, for linear systems, asymptotic and exponential stability are equivalent.

European School for Advanced Studies in Reduction of Seismic Risk
(ROSE School)
University of Pavia
Pavia 27100, Italy
(K.C.M.)

Department of Civil Engineering
University of Patras
GR–26500 Patras, Greece
(A.S.P.)

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